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AI-Powered Smart Parking System

نظام مواقف السيارات الذكي المدعوم بالذكاء الاصطناعي

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Declaration

We, the undersigned, hereby declare that the work presented in this report titled "**AI-Powered Smart Parking System**" is the result of our collective efforts and original research. The project was carried out under the supervision of Dr. Ahmed Moawad in partial fulfillment of the requirements for the degree of Artificial Intelligence. We confirm that this work has not been submitted elsewhere for any degree or qualification and that it complies with the ethical and academic standards of Egyptian Russian University.

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List of Abbreviations

- **AI** - Artificial Intelligence
- **IoT** - Internet of Things
- **LPR** - License Plate Recognition
- **ANPR** - Automatic Number Plate Recognition
- **CNN** - Convolutional Neural Network
- **CV** - Computer Vision
- **ESPS** - Equitable Smart Parking System
- **OCR** - Optical Character Recognition
- **YOLO** - You Only Look Once (Object Detection Algorithm)
- **ML** - Machine Learning
- **API** - Application Programming Interface
- **GPS** - Global Positioning System
- **UI** - User Interface
- **UX** - User Experience
- **SaaS** - Software-as-a-Service
- **AES** - Advanced Encryption Standard
- **SSL** - Secure Sockets Layer
- **2FA** - Two-Factor Authentication
- **OTP** - One-Time Password
- **ROI** - Return on Investment
- **TCO** - Total Cost of Ownership
- **PII** - Personally Identifiable Information
- **GDPR** - General Data Protection Regulation
- **LSTM** - Long Short-Term Memory
- **DQN** - Deep Q-Network
- **AHP** - Analytic Hierarchy Process
- **NFC** - Near Field Communication
- **OpenCV** - Open Source Computer Vision Librar

ABSTRACT

The rapid growth of urban areas has led to a significant increase in the number of vehicles, resulting in severe parking challenges such as traffic congestion, wasted time, and increased fuel consumption. This project aims to develop a fully autonomous, AI-powered smart parking system that enhances administrative efficiency and vehicle security. The core of the proposed methodology is a License Plate Recognition (LPR) system powered by a custom-trained Convolutional Neural Network (CNN) to ensure high-accuracy vehicle identification. This deep learning module, inspired by foundational architectures such as AlexNet, is integrated with an IoT-based sensor network and a mobile application that enables user reservations and payments, thereby automating the entire parking process. The expected outcomes include a fully automated parking cycle, a centralized backend system for management and analytics, and real-time data accessibility for drivers. This system provides a secure and streamlined user experience from arrival to departure without the need for human intervention. Ultimately, the proposed solution significantly reduces the time drivers spend searching for parking spaces, contributing to the mitigation of urban traffic congestion and the reduction of fuel consumption and environmental impact.

CHAPTER 1 INTRODUCTION

1.1 Overview

This project aims to develop an AI-powered smart parking system that automates and optimizes the parking process using artificial intelligence, computer vision, and IoT technologies. The system is designed to efficiently manage parking spaces, reduce traffic congestion, and provide a smooth and secure user experience.

It operates by detecting vacant spots in real time through smart sensors and cameras, identifying vehicles automatically using Number Plate Recognition, and updating availability through a connected IoT platform. A dedicated mobile application allows users to find, reserve, and pay for parking spaces seamlessly.

This project provides a cost-effective solution that can be set up in malls, airports, universities, and urban parking facilities. The system not only relieves traffic congestion but also allows for smart city development by increasing urban mobility and preserving the environment.

The proposed solution minimizes human intervention, enhances security, and contributes to reducing fuel consumption and emissions caused by prolonged parking searches. Overall, the system represents a step toward smarter, more sustainable urban mobility and efficient parking management.

1.2 Problem Description

In modern cities, finding a suitable parking has become a daily challenge faced by most drivers. The main reason lies in the absence of smart parking management systems, which leads to wasted time, increased traffic congestion, higher fuel consumption, and rising stress levels among drivers. Traditional systems still rely on manual monitoring, payment, and access control, making them slow, inaccurate, and difficult to scale with the growing number of vehicles. The lack of License Plate Recognition (LPR) system causes long queues and delays at entrances, while drivers spend a significant amount of time searching for vacant spots, especially during peak hours, due to the absence of real-time updates or pre-booking options. Furthermore, manual or cash payments result in additional delays and errors, with no effective system to track vehicles or ensure optimal space utilization. Pricing systems are fixed and lack transparency, while management suffers from the absence of a centralized dashboard and analytical reports that support decision-making and performance improvement. In addition, users face difficulties caused by complex application interfaces and limited integration with

navigation tools, whereas systems that rely entirely on internet connectivity become non-functional when the connection is lost.

1.3 Project Motivation

In the heart of bustling urban centers, the seemingly simple task of finding a parking spot has evolved into a significant daily challenge. Traditional parking systems are plagued by inefficiencies that extend far beyond minor inconveniences. Drivers often waste a substantial amount of time and effort searching for vacant spaces, a process that can consume up to an estimated thirty percent of their total journey time in crowded areas [1]. This futile search directly contributes to the exacerbation of traffic congestion, increased fuel consumption, and a rise in environmentally harmful carbon emissions. Furthermore, the lack of effective monitoring and security mechanisms in conventional lots leaves vehicles vulnerable to theft and damage, while manual management methods are characterized by low operational efficiency and a heavy reliance on human personnel.

It is this multifaceted problem that fuels our motivation to develop the AI-powered Smart Parking System. The project is driven by the vision of leveraging cutting-edge technology to transform the parking experience radically. As students of artificial intelligence, we are compelled to apply our academic knowledge to create practical solutions for real-world issues. We envision a seamless system that integrates computer vision, the Internet of Things (IoT), and mobile applications to provide a fully automated and intelligent parking management solution.

Our motivation is not just to build a new technology, but to engineer a solution with a tangible, positive impact. The primary goal is to enhance operational efficiency, reduce traffic, and provide a secure and user-friendly experience for drivers. By automating vehicle identification, providing real-time data on vacant spots via a dedicated mobile app, and minimizing the reliance on human intervention, our system aims to give valuable time back to drivers. Ultimately, this project is driven by the opportunity to contribute to smarter, more sustainable cities by reducing traffic congestion and its associated environmental footprint.

1.4 Project Objectives.

Every project should start with clear objectives that define its purpose

and direction. These objectives help keep development focused and ensure an effective final system. The AI-powered Smart Parking System is guided by the following objectives:

- Automate the parking management process using artificial intelligence to minimize human intervention and errors.
- Detect and monitor available parking spaces in real time through the use of cameras and computer vision algorithms.
- Provide an intelligent parking guidance system that directs drivers to the nearest available spot efficiently.
- Reduce traffic congestion within parking areas by optimizing vehicle flow and minimizing search time for parking spaces.
- Enhance user experience through a mobile or this project-based application that displays live parking availability and navigation support.
- Collect and analyze parking data to generate insights for future optimization and smart city integration.
- Promote sustainability by reducing fuel consumption and carbon emissions caused by unnecessary vehicle idling and searching.

CHAPTER 2 RELATED WORK

2.1 Literature Review

Tarteel Abdelfattah Elattar .et al [1] sought to extend LPR systems to include motorcycle plates, which are often neglected but increasingly relevant in crowded cities. They aimed to achieve accurate detection and recognition for both cars and motorcycles. Methodology A dataset of over 1,200 labeled images was

created. YOLOv8 was used for detection, followed by OCR for recognition. The system achieved 100% plate detection accuracy but only 80% recognition accuracy for motorcycles, reflecting challenges in image clarity and size. Strengths Unique dataset covering both vehicle and motorcycle plates. Applied and tested on video streams for real-world simulation. Strong detection performance even with low-resolution cameras. Limitations Recognition accuracy decreases sharply for motorcycles. No integration with any parking control or management framework. Dataset size too small for large-scale generalization. Research Gap Elattar et al. highlighted the difficulty of handling diverse vehicle types but did not address how recognition links to automated parking workflows. This project builds on this by incorporating vehicle type recognition as part of the entry validation system, allowing differentiated slot allocation and pricing. This project also aims to expand the dataset for higher recognition accuracy under real parking conditions.

Kalaivani et al. [2] aimed to design a cost-effective IoT system that automates parking occupancy detection and minimizes driver waiting time through sensor-based data collection. Methodology They used ultrasonic sensors connected to Arduino microcontrollers to detect whether parking slots were occupied. Slot status was displayed on an LCD monitor and updated in real time. The system also controlled gate access using servo motors. Strengths Affordable, easy-to-implement prototype. Real-time operation with basic automation. Demonstrates IoT's potential in improving parking logistics. Limitations Lacks scalability for large parking facilities. No AI integration — slot allocation and user guidance remain manual. No vision component, vehicle identification, or security measures. Research Gap While Kalaivani et al. introduced a practical IoT prototype, their system remains low-level, focusing on hardware without intelligence. This project advances this by integrating AI-powered vision (for recognition and monitoring) with IoT sensors (for slot status and environment tracking), coordinated through a centralized backend and user-facing mobile app — combining cost efficiency with automation and intelligence.

Sajid Rehman et al. [3] investigate the integration of AI-driven slot detection with IoT payment solutions. The primary objective is to enhance parking efficiency, improve the user experience, and reduce urban traffic congestion. Methodology The system's approach is multi-faceted, combining several key technologies: AI-Based Image Recognition: Utilizes cameras and machine learning algorithms to visually analyze parking areas and accurately determine in real-time whether a slot is occupied or vacant. IoT-Enabled Sensors: Complements the camera system with physical sensors to provide a redundant layer of data, ensuring high reliability in occupancy detection. Contactless Payment Integration: Incorporates modern payment gateways, allowing for seamless and automated billing through mobile applications, NFC, or other tap-to-pay methods. Strengths Accurate Real-Time Detection: The use of AI with image recognition delivers highly precise and instantaneous information on parking availability, providing

reliable data for both drivers and operators. Streamlined Automated Billing: The system eliminates the need for manual payment kiosks or cash transactions, resulting in a fast, convenient, and frictionless billing process for users. Limitations The system faces two major challenges. First, it is vulnerable to significant cybersecurity risks, as it handles sensitive payment and personal data that could be targeted by attackers. Second, the high infrastructure cost of installing advanced cameras, sensors, and the required AI processing hardware can be a major barrier to widespread implementation. Research Gap The study identifies a critical need for standardized regulatory frameworks to govern the operation, security, and data sharing of such systems. Additionally, a gap exists in the development of improved data privacy measures to ensure user information is protected, which is essential for gaining public trust and encouraging adoption.

Mert Ozkaya et al. [4] proposed a standardized architecture that defines how smart parking systems can be structured to ensure modularity, scalability, and interoperability between IoT, cloud, and mobile components. Methodology The authors conducted an analytical review of multiple existing systems and proposed a reference framework consisting of layered components: sensing, communication, processing, storage, and application layers. The work emphasizes interoperability, cloud integration, and performance monitoring.

Strengths Provides a blueprint architecture for smart parking systems. Emphasizes integration, modularity, and scalability. Highlights key design challenges and performance considerations. Limitations Theoretical only; no implementation or experimental validation. Lacks specific algorithms for detection, guidance, or payment automation. Research Gap Ozkaya et al. provide a useful theoretical structure but no implementation proof. This project brings architecture to life — translating theory into a working prototype that combines AI-based detection, IoT communication, and a mobile interface, effectively validating the architecture through practical deployment.

Nadir Ali et al. [5] integrated AI and IoT for a smart parking system that offers real-time space monitoring and automated billing, aiming to boost urban efficiency and sustainability. Methodology The system uses IoT sensors in each parking spot to detect vehicle presence. This data is processed on a cloud platform where AI algorithms manage space allocation, guide drivers via a mobile app, and handle automated billing using license plate recognition. Strengths Efficiency: Optimizes parking space usage and reduces manual oversight. Real-Time Data: Provides live parking availability to drivers, reducing traffic congestion. Sustainability: Lowers fuel consumption and emissions by minimizing search times. User Experience: Offers a seamless, automated payment and parking process. Limitations The system's reliance on connected devices and sensitive data requires robust cybersecurity measures to protect user information and payment details from potential threats. Research Gap A key gap identified is the need for improved integration and interoperability frameworks. This would allow the

system to connect seamlessly with other smart city platforms (like traffic management and public payment systems) for effective large-scale deployment.

Zeeshan Haider et al. [6] developed a fully automated, IoT-based parking system focused on providing dynamic slot allocation and secure, automated payment processing to maximize efficiency. Methodology The system was built using a network of sensors for real-time slot detection. Data from these sensors is transmitted to a central cloud platform for processing. This platform manages a live map of available spots and communicates it to users. A key feature is the use of a blockchain-secured payment system, ensuring that all transactions are transparent, immutable, and secure. Strengths Improved Slot Utilization: The system dynamically assigns spots, ensuring that available spaces are filled quickly and efficiently. Real-Time Updates: Drivers receive instant and accurate information about parking availability, which significantly reduces search time and traffic. Enhanced Payment Transparency: By leveraging blockchain technology, the system offers a highly secure and transparent billing process, eliminating disputes and preventing fraud. Limitations The primary drawbacks are the high initial infrastructure cost associated with deploying sensors, gateways, and cloud services, along with persistent data security challenges in protecting the entire IoT network from cyber threats. Research Gap The study identifies a clear need for the development of scalable and more cost-effective IoT frameworks. Furthermore, there is a critical gap in creating solutions that incorporate strong privacy protection by default, ensuring that user location and behavioral data are handled securely and ethically.

Liang and Zhang [7] Developed an IoT-based intelligent parking system to enable real-time detection, visualization, and reservation of spaces. Methodology Used ultrasonic sensors with CC2530 microcontrollers and ZigBee for communication. Data transmitted to OneNet cloud, with a WeChat applet for users and web-based management for admins. Included e-payment features. Strengths ZigBee offers low-power, efficient networking; user-friendly WeChat integration. Limitation High hardware cost for per-slot sensors, and lack of real-world performance data. Research Gap Need to balance between per-slot sensor accuracy and vision-based system scalability.

Amany Sarhan et al. [8] aimed to build a reliable license plate recognition system adapted to Egyptian plate formats, which are bilingual (Arabic-English) and prone to unique environmental and visibility challenges. Methodology The authors employed YOLOv8 for license plate detection, and compared Easy-OCR with a custom CNN model for character recognition. Preprocessing steps like skew correction and binarization were applied. The model achieved up to 99% recognition accuracy for English plates and slightly less for Arabic ones. Strengths First to explicitly address Egyptian license plates. Dual recognition methods (OCR + CNN) improved robustness. Strong preprocessing pipeline that enhances image clarity and segmentation. Limitations Scope limited to recognition only; no parking

guidance or management system. Focused purely on computer vision; lacks IoT or real-time user interaction. No discussion of integration with hardware systems (e.g., gates, sensors). Research Gap While Sarhan et al. successfully localized ALPR for Egypt, the study lacks a broader operational context. This project will extend this localized recognition capability into a fully integrated AI-driven parking system, using Egyptian plate recognition for vehicle authentication, while IoT modules and a mobile app automate slot allocation, payment, and exit — bridging the gap between vision research and real-world parking management.

Mejdl Safran et al. [9] aimed to enhance the accuracy and reliability of automatic license plate recognition (ALPR) within smart parking systems. Their goal was to design a high-precision vision-based system that minimizes dependency on additional sensors and performs efficiently in real-time scenarios. Methodology The researchers developed a multi-stage recognition pipeline: YOLOv5 was used for detecting the license plate area, YOLOv8 for identifying individual characters, and a custom CNN for character recognition. The dataset consisted of real Saudi Arabian license plates captured under different lighting conditions. Their system achieved a recognition accuracy exceeding 96%, significantly outperforming single-stage models. Strengths Multi-stage pipeline improves precision and reduces false recognition rates. Real-time inference suitable for deployment in parking environments. Includes a visualization dashboard for real-time monitoring and analytics. Limitations Dataset and performance are tailored to Saudi plates only. Limited generalization to other countries or varied environmental conditions. The system handles recognition but not parking space detection, user guidance, or payment automation. Research Gap While Safran et al. focused mainly on improving ALPR accuracy, their system stops short of full parking management. This project expands on this by integrating the recognition module within a complete AI-powered ecosystem — combining IoT sensors for slot detection, a mobile app for user interaction and payment, and backend automation for management. This transforms a detection model into a functional, end-to-end parking system.

Abdelrahman Osman Elfaki [10] proposed a dynamic parking control system that allows real-time slot monitoring and allocation to minimize congestion and manual supervision. Methodology The system used IoT sensors and cameras to detect occupancy, combined with a mobile application that displayed real-time parking availability. A prototype implementation demonstrated efficient space utilization and reduced search time. Strengths Functional prototype integrating IoT and mobile components. Demonstrated practical improvement in parking management. Real-time monitoring enhanced user convenience. Limitations Limited to small organizational setups. Did not employ AI or computer vision for vehicle recognition. Lacked scalability for public or multi-level parking environments. Research Gap Elfaki et al. succeeded in IoT-based monitoring but lacked AI-powered recognition or secure digital integration. This project advances

this by combining AI-based ANPR with IoT sensors, enabling automated entry/exit, user payments, and scalable multi-lot management.

Qin et al. [11] integrated decision-making methodology for an Autonomous Parking System (APS) focusing on two previously underserved modules: optimal parking space selection and vehicle driving decisions. The overall aim is to improve the efficiency and smoothness of autonomous vehicle driving while ensuring safety. Methodology The work is structured around two main decision aspects: Optimal Parking Space Selection: A multi-attribute decision method is used, combining the Analytic Hierarchy Process (AHP) (for subjective factors like driver preference) and the Entropy Method (for objective factors). The final selection uses TOPSIS. Factors include parking space type,walking distance, and driving distance. Vehicle Driving Decisions: A novel hybrid method combining a rule based Behavior Tree with a learning-based Deep Q-Learning Network (DQN) is used. This decision process incorporates a novel model for target vehicle behavior prediction (LSTM) and trajectory prediction (Constant Velocity or Five-degree polynomial) in specific parking lot scenarios. Strengths Holistic Decision-Making: Addresses the entire decision module (selection + dynamic driving behavior) which is largely overlooked in current APS research focusing on perception and control. Integrated Multi-Attribute Model: Overcomes the bias of previous work by integrating both subjective and objective factors for space selection, leading to a more reasonable decision. Safety and Efficiency Fusion: The Behavior Tree/DQN/Prediction hybrid improves both safety (obstacle avoidance) and efficiency (smoother driving, higher average speed) in complex parking scenarios. Limitations The rule-based Behavior Tree method cannot traverse and make decisions for all possible scenarios due to the inherent limitations of rule based systems. Future work is needed to incorporate a learning-based approach into the upper-level decision-making to adapt to wider range of complex parking lot situations. research gap is the lack of a unified framework for multi-criteria optimal parking selection and safe, adaptive dynamic driving in complex parking lots. This project can address this by building on the hybrid Behavior Tree/DQN model (Qin et al., 2024), specifically by integrating more learning-based methods into its high-level decision-making.

Jemmali et al. [12] creates a smart parking allocation system that equitably distributes *people* (not just vehicles) to minimize the gap in the number of people between parking spaces and reduce human congestion. Methodology The problem was formulated as an NP-hard problem. The authors developed seven new heuristic algorithms to solve it, testing them across 2430 cases. Key Finding the MR algorithm was the most effective, finding the best solution in 96.1% of instances with a minimal average gap (0.02) and a fast execution time (0.007s), surpassing the previous best algorithm (C3S). Strengths The primary strength is the innovative focus on distributing people, which directly addresses congestion

at entrances/exits. The MR algorithm is efficient, and the system is designed for dynamic, real-time allocation. Limitations The algorithms are heuristics (approximate solutions). Future work could use meta-heuristics to find exact solutions. Identified Research Gap The key gap in literature is the failure of existing systems to address the *human factor*. Prior methods focus on optimizing *vehicle* allocation (based on proximity, cost, or vehicle count), which ignores the varying number of passengers per car. This leads to an inequitable distribution of people, causing "unwelcome chaos" and congestion.

Ref No.	Year	Technique	Strengths	Limitations
[1]	2025	Using a dataset of over 1,200 images, the system employed YOLOv8 for detection and OCR for recognition. It achieved 100% plate detection accuracy, but motorcycle recognition was only 80% accurate due to poor image clarity and size.	• Unique dataset with both car and motorcycle plates. • Tested on real-world video streams. • High detection accuracy, even with low-resolution cameras.	• Low recognition accuracy for motorcycles. • Not integrated with a parking management system. • The dataset is too small for large-scale use.
[2]	2025	Used ultrasonic sensors with an Arduino to detect parking slot occupancy. The status was displayed in real-time on an LCD, and the system-controlled gate access with	• Affordable and easy-to-implement prototype. • Offers real-time operation and basic automation.	• Not scalable for large parking facilities. • Lacks AI for user guidance or slot allocation.

		servo motors.	• Demonstrates the potential of IoT in parking.	• Has no vehicle identification or security features.
[3]	2025	This multi-faceted system uses AI-powered cameras for visual, real-time occupancy detection. It's complemented by IoT sensors for higher reliability and includes contactless payment integration for automated billing.	• Accurate, real-time AI occupancy detection. • Convenient automated billing.	• High cybersecurity risks (payment/personal data). • High infrastructure and hardware costs.
[4]	2025	the authors proposed a layered reference framework (sensing, communication, processing, storage, and application). The framework emphasizes interoperability, cloud integration, and performance monitoring.	• Provides a smart parking architectural blueprint. • Emphasizes integration, modularity, and scalability. • Highlights design and performance challenges.	• Purely theoretical; no implementation or validation. • Lacks specific algorithms (e.g., detection, guidance, payment).

[5]	2025	The system uses IoT sensors in each parking spot to detect vehicle presence. This data is processed on a cloud platform where AI algorithms manage space allocation, guide drivers via a mobile app, and handle automated billing using license plate recognition.	• Optimizes space and reduces manual oversight. • Real-time data reduces traffic. • Lowers emissions by reducing search time. • Smooth, automated parking and payment.	• Requires strong cybersecurity for sensitive user/payment data.
[6]	2025	A sensor network detects real-time slot occupancy, sending data to a central cloud platform for processing. The platform provides users with a live map of available spots and utilizes a blockchain-secured system for transparent and secure payment transactions.	• Dynamic slot assignment improves space utilization. • Real-time updates reduce search time and traffic. • Blockchain provides secure, transparent, fraud-proof billing.	• High initial infrastructure costs (sensors, gateways, cloud). • Significant data security challenges for the IoT network.
[7]	2024	Used ultrasonic sensors with CC2530 microcontrollers and ZigBee for communication. Data transmitted to OneNet cloud, with a WeChat applet for users and web-based management for	• Low-power, efficient ZigBee networking. • User-friendly WeChat integration.	• High hardware cost for individual slot sensors. • Lack of real-world performance data.

		admins. Included e-payment features.		
[8]	2024	Using YOLOv8 for detection and a custom CNN for recognition (compared against Easy-OCR), the system applied preprocessing like skew correction. It achieved up to 99% recognition accuracy for English plates, with slightly lower results for Arabic plates.	• First system for Egyptian license plates. • Robust dual recognition (OCR + CNN). • Strong preprocessing pipeline for image clarity.	• Scope limited to LPR; no parking guidance or management. • Lacks IoT integration or real-time user interaction. • Does not address hardware integration (gates, sensors).
[9]	2024	Using a multi-stage pipeline, researchers employed YOLOv5 for plate detection, YOLOv8 for character identification, and a custom CNN for character recognition. The system was tested on a dataset of Saudi Arabian license plates under various	• Multi-stage pipeline enhances precision. • Capable of real-time inference. • Includes a dashboard for real-time monitoring.	• Tailored specifically to Saudi Arabian license plates. • Limited generalization to other countries/conditions. • Lacks parking space detection,

		lighting conditions and achieved over 96% recognition accuracy, outperforming single-stage models.		user guidance, or payment features.
[10]	2024	The system used IoT sensors and cameras to detect occupancy, combined with a mobile application that displayed real-time parking availability. A prototype implementation demonstrated efficient space utilization and reduced search time.	• Functional prototype with integrated IoT and mobile components. • Demonstrated practical improvements in parking management. • Real-time monitoring for enhanced user convenience.	• Limited to small-scale organizational setups. • Lacks AI or computer vision for vehicle recognition. • Not scalable for public or multi-level parking environments.

[11]	2024	The system uses a multi-attribute model (AHP, Entropy, TOPSIS) to select the optimal parking spot. For vehicle navigation, it employs a hybrid system combining a rule-based Behavior Tree with a learning-based Deep Q-Network (DQN), using an LSTM to predict the behavior of other vehicles.	• Holistic Decision-Making: Covers the entire process from selection to driving. • Integrated Multi-Attribute Model: Combines subjective and objective factors for selection. • Safety and Efficiency Fusion: Hybrid approach improves both in complex scenarios.	• Rule-based Behavior Tree cannot cover all possible scenarios. • Needs a learning-based approach to adapt to more complex situations.
[12]	2022	The system used IoT sensors and cameras to detect occupancy, feeding real-time availability data to a mobile application for users.	• Delivered a functional prototype integrating IoT and a mobile app. • Showed practical improvements in parking management.	• Only suitable for small-scale organizational setups. • Lacks AI or computer vision for vehicle recognition

CHAPTER 3 PROJECT ANALYSIS

3.1 Problem Statement

Current parking management systems in Egypt and many smart city projects still suffer from inefficiency, congestion, and the absence of intelligent automation. Most existing solutions rely on basic magnetic or ultrasonic IoT sensors, which provide only binary occupancy data (occupied/free) without contextual awareness or visual validation. This lack of computer vision integration, such as License Plate Recognition (LPR), limits the system's ability to perform secure, real-time vehicle identification and automated slot management. As a result, parking operations remain partially manual, prone to human error, and vulnerable to data manipulation or unauthorized access due to weak IoT network security.

3.2 Problem Analysis

A deeper analysis of existing research and prototypes (as reviewed in Chapter 2) reveals that the current landscape is technically fragmented and lacks a unified, scalable design: Fragmented Research Direction: Existing systems are either Vision-based (LPR) with limited IoT connectivity, or IoT-sensor-based solutions without real-time computer vision for secure vehicle validation. Limited Integration Layer: Few systems successfully merge AI algorithms with IoT frameworks in a unified architecture capable of real-time decision-making and data synchronization. Scalability Issues: Most prototypes are tested in small, controlled environments and cannot scale to large, multi-level parking infrastructures. Cyber security Gaps: Communication between IoT nodes and cloud platforms often lacks encryption and authentication, exposing the system to spoofing and data interception. Performance Constraints: Sensor-only systems generate binary data, which is insufficient for intelligent slot allocation or anomaly detection. Operational Inefficiency: Without AI-driven analytics, user guidance and payment automation remain slow and fragmented. Hence, the core problem lies in the absence of an integrated IoT–AI framework that combines sensor data, computer vision, and cybersecurity mechanisms to deliver a reliable and scalable smart parking solution.

3.3 Project Visibility Study

3.3.1 Executive Summary

The escalating vehicular congestion in urban centers and large-scale facilities necessitates a paradigm shift from conventional parking management to intelligent, data-driven systems. This study assesses the viability of the Equitable Smart Parking System (ESPS), a novel solution designed not merely to count vehicles, but to manage human pedestrian load. The core problem identified is the inequity of human load distribution resulting from vehicle-centric allocation models. ESPS solves this by integrating Computer Vision (CV) technology with advanced Heuristic Algorithms to allocate parking based on the number of occupants per vehicle. Key Findings: The project is demonstrably technically feasible due to the maturity of CV and IoT technologies, economically viable given the substantial return on investment (ROI) derived from efficiency gains, and operationally superior to all current market alternatives. Recommendation: The implementation of the ESPS prototype is strongly recommended. Its execution will yield a robust, novel, and academically significant outcome, positioning the system as a critical piece of next-generation Smart City infrastructure. Smart city platforms (like traffic management and public payment systems) for effective large-scale deployment.

3.3.2 Product/Service Description and Project Goals

3.3.2.1 Project Definition and Problem Statement

The fundamental flaw in current smart parking systems is their reliance on the vehicle count as the primary load-balancing metric. A four-seater vehicle with one occupant is treated identically to one with five occupants. This results in the uneven concentration of people (pedestrian load) in certain parking zones, leading to unnecessary bottlenecks, extended egress/ingress times, and diminished user experience at critical infrastructure points (e.g., elevators, payment kiosks, entry gates). The ESPS resolves this by establishing equitable allocation as the primary objective, minimizing the Gap Value (the difference in total human load between parking zones).

3.3.2.2 Project Goals

Category	Primary Goal	Critical Success Metric

Core	Achieve Equitable Human Load Distribution across all available parking zones.	Reduce the average Gap Value (as defined by Jemmali et al., 2022) by over 90% compared to traditional FIFO systems.
Operational	Significantly reduce driver search time and internal facility congestion.	Decrease the average time taken from facility entry to parking slot placement by 30%.
Technical	Accurately and instantaneously determine vehicle occupancy upon entry.	Achieve occupancy detection accuracy of 95% or higher in real-time under varying light conditions.

3.3.3 Technology and Architectural Considerations

3.3.3.1 Core Technologies

Component	Technology Utilized	Strategic Rationale
Occupancy Detection	Deep Learning Models (e.g., YOLO or R-CNN) via Computer Vision and high-resolution cameras.	Enables fast, non-intrusive estimation of passenger count, which is the foundational input for the system.
Allocation Engine	NP-Hard Heuristic Algorithms (e.g., the MR Algorithm) combined with real-time IoT data feeds.	Guarantees an optimal or near-optimal solution for load balancing within the milliseconds required for dynamic allocation (high performance and low latency).
Data Backbone	Edge Computing Architecture combined with 5G/Wi-Fi Mesh.	Allows for instantaneous data processing at the "edge" (near the cameras), minimizing network latency and ensuring rapid decision-making.

3.3.3.2 Intellectual Property (IP) and Data Privacy

These considerations are critical for any academic and commercial deployment:

- Intellectual Property Protection: The ESPS's core innovation lies in the adaptation and parameterization of the allocation algorithm for real-world equity objectives. The specific mathematical formulation and software implementation will be documented as the project's novel IP. All background research (e.g., the work of Jemmali et al.) will be fully cited.
- Data Privacy & Personal Data Handling (GDPR/Data Protection Compliance): The system must be designed from the ground up on the principle of Anonymization. The CV unit will perform its detection and immediately convert the image data into a single, non-personal numeric value (the occupancy count). The raw video/image data that allows for personal identification will be deleted or purged immediately after processing. No personally identifiable information (PII) is collected, stored, or processed, mitigating the most significant legal and ethical risk associated with video surveillance. user-friendly WeChat integration. Limitation High hardware cost for per-slot sensors, and lack of real-world performance data. Research Gap Need to balance between per-slot sensor accuracy and vision-based system scalability.

3.3.4 Product/Service Marketplace Analysis

3.3.4.1 Target Region and Customer Need

Target Region: The initial deployment will focus on a region characterized by high, variable pedestrian traffic and constrained parking capacity, such as a major University Campus or a Central Business District (CBD) parking garage. Market Need (Investment Justification): There is a pronounced, underserved need for an "equity-focused" parking solution. Current investment in smart parking is justified only by efficiency, not by improved user experience or safety. The ESPS shifts this justification: Investment in ESPS is an investment in crowd management, operational safety, and customer satisfaction, making the ROI multi-faceted.

3.3.4.2 Competitor Analysis and Market Demand

Competitor Type	Core Solution Offered	Competitive Weakness / Research Gap	ESPS Unique Value Proposition
Traditional Systems	Manual attendance, basic sensors for slot occupancy (RED/GREEN).	Zero optimization, massive delays, and no data utilization.	Immediate, algorithmically optimized allocation.
Current Smart Systems	Simple shortest- path routing; vehicle-count-based allocation.	Ignores Human Load: Fails to prevent pedestrian bottlenecks and congestion at key access points.	The Equity Factor: Guarantees a balanced and seamless experience for *all* occupants, not just the vehicle driver.

3.3.5 Marketing Strategy

The marketing and communication strategy for ESPS will be centered on its unique selling proposition: Equity and Flow.

- Focus Message: "ESPS: Optimized for People, Not Just Vehicles. The Future of Urban Flow."
- Targeting: Directly approach facility operators and city planners responsible for crowd management and public safety, not just IT or transportation departments.
- Proof of Concept: Utilize the final graduation project documentation (including the measurable reduction in Gap Value and search time) as the primary Case Study for convincing future adopters.

3.3.6 Organization, Staffing, and Schedule

3.3.6.1 Organization and Staffing

The successful implementation of the prototype requires a

focused, cross-disciplinary team:

Role	Key Expertise Required
System Architect (Project Lead)	Integration, system modeling, financial analysis (Oversees the entire Feasibility and Implementation).
Algorithm Developer	Non-convex optimization, Heuristic programming, Python/C++ development (Core engine coding).
Vision & Hardware Engineer	Computer Vision (TensorFlow/PyTorch), Camera/Sensor integration, Edge Computing deployment.

3.3.7 Financial Projections and Risk Management

3.3.7.1 Investment Justification and Gains

The economic case is robust, justifying investment through enhanced operational efficiency:

Investment Gain Category	Description of Profit/Value	Calculation of Return
Revenue Optimization	Increased total parking throughput due to faster turnover and reduced bottlenecks (estimated 15% increase).	Direct revenue increase from parking fees.
Cost Avoidance	Reduced staffing needs (less need for manual traffic control) and lower maintenance costs due to decreased congestion-related wear and tear.	Reduction in Total Cost of Ownership (TCO) over a five-year period.
Intangible Value	Enhanced user experience, positive facility reputation, and alignment with municipal Smart City initiatives.	Increased commercial attractiveness and higher customer loyalty.

3.3.7.2 Risk Assessment and Mitigation Strategies

A thorough analysis identifies potential "Red Flags" and necessary pre-emptive mitigation actions,

ensuring due diligence for the graduation committee.

Risk Category	Key Risk / Red Flag	Mitigation Strategy
Technical/Operational	Sensor Reliability: CV failure due to extreme weather, low light, or sensor calibration drift.	Implement Sensor Fusion (e.g., using weight sensors or induction loops as a secondary data source) and utilize Edge Processing for robust, immediate fault detection.
Financial	Hardware Cost Overruns: High cost of procuring robust, industrial-grade CV cameras and processors.	Adopt an initial Software-as-a- Service (SaaS) model where possible, leveraging existing client CCTV infrastructure and focusing expenditure only on the core computing unit.
Market/Planning	Regulatory/Bureaucrati c Inertia: Slowness in adoption by city/facility management due to reliance on legacy systems.	Focus the pilot on a single, private facility (e.g., a mall) or a university campus where the decision-making chain is shorter, providing undeniable proof of concept metrics.

3.3.8 Findings and Recommendations

3.3.8.1 Findings

The Equitable Smart Parking System (ESPS) project is not

only viable but presents a compelling case for execution. It successfully addresses a critical gap in existing literature and commercial technology by shifting the optimization target from *vehicles* to *people*.
The project exhibits:

1. High Technical Fidelity: Achievable with current, mature technologies (CV, Heuristics).
2. Strong Economic Justification: The gains in efficiency and operational flow far outweigh the development and deployment costs.
3. Unique Competitive Position: The equity-based allocation ensures long-term differentiation and market leadership in sophisticated smart city applications.

3.3.8.2 Recommendation

It is unequivocally recommended that the student proceeds with the development and deployment of the Equitable Smart Parking System (ESPS) prototype.

Business Model Canvas

Key Partners

- Academic & Advisory:Our supervisor (Dr. Ahmed Moawad) and university (Egyptian Russian University).
- (Hypothetical) Business Partners:
- Hardware Suppliers: Manufacturers of cameras, IoT sensors, and smart gates.
- Cloud Service Providers: For hosting the backend system, database, and AI models.
- Payment Gateway Providers: To process the electronic payments from the mobile app.
- Launch Partners: The first facilities (mall, university, airport) that agree to pilot or install the system

Key Activities

- Developing the custom ANPR model using CNN and Computer Vision.
- Building the mobile app and backend management system.
- Integrating AI software with IoT sensors and hardware (gates, cameras)

Key Resources

- The custom AI/CNN model for ANPR.
- Physical hardware (cameras, IoT sensors, smart gates).
- The development team and supervisor, Dr. Ahmed Moawad

Value Propositions

- For Drivers (End-Users):
- Time & Cost Savings: Reduces time, fuel, and stress from parking searches.
- Seamless Experience: Fully automated and secure process, from reservation to payment.
- Real-Time Information: Live data on vacant spots via a mobile app.
- Enhanced Security: Improved vehicle monitoring and protection.
- For Facility Owners (B2B):
- Operational Efficiency: Automates parking, boosts administrative efficiency, and reduces staff reliance.
- Data & Analytics: Central control panel with reports and analytics for decision-making.
- Optimized Space: Efficiently manages and maximizes space utilization.

Customer Relationship

- Automated & Self-Service: For drivers, via the app and automated ANPR/gates.
- Dedicated Assistance: For owners, including installation, training, and support.

Channels

- Mobile Application: For drivers to access the service (reservations, payments).
- Website & Dashboard: A website for user marketing and a dashboard for partner analytics.
- Partnerships with Municipalities & Smart City Programs: Government partnerships to build trust and accelerate expansion.
- Direct Sales/Installation: To sell and install the system for facility owners

Customer Segments

- B2C (Business-to-Consumer) - The End-Users:
- Drivers in crowded urban areas who waste time and fuel searching for parking.
- B2B (Business-to-Business) The Facility Owners:
- Parking facility operators, such as those at malls, airports, universities, and other urban facilities.

Cost Structure

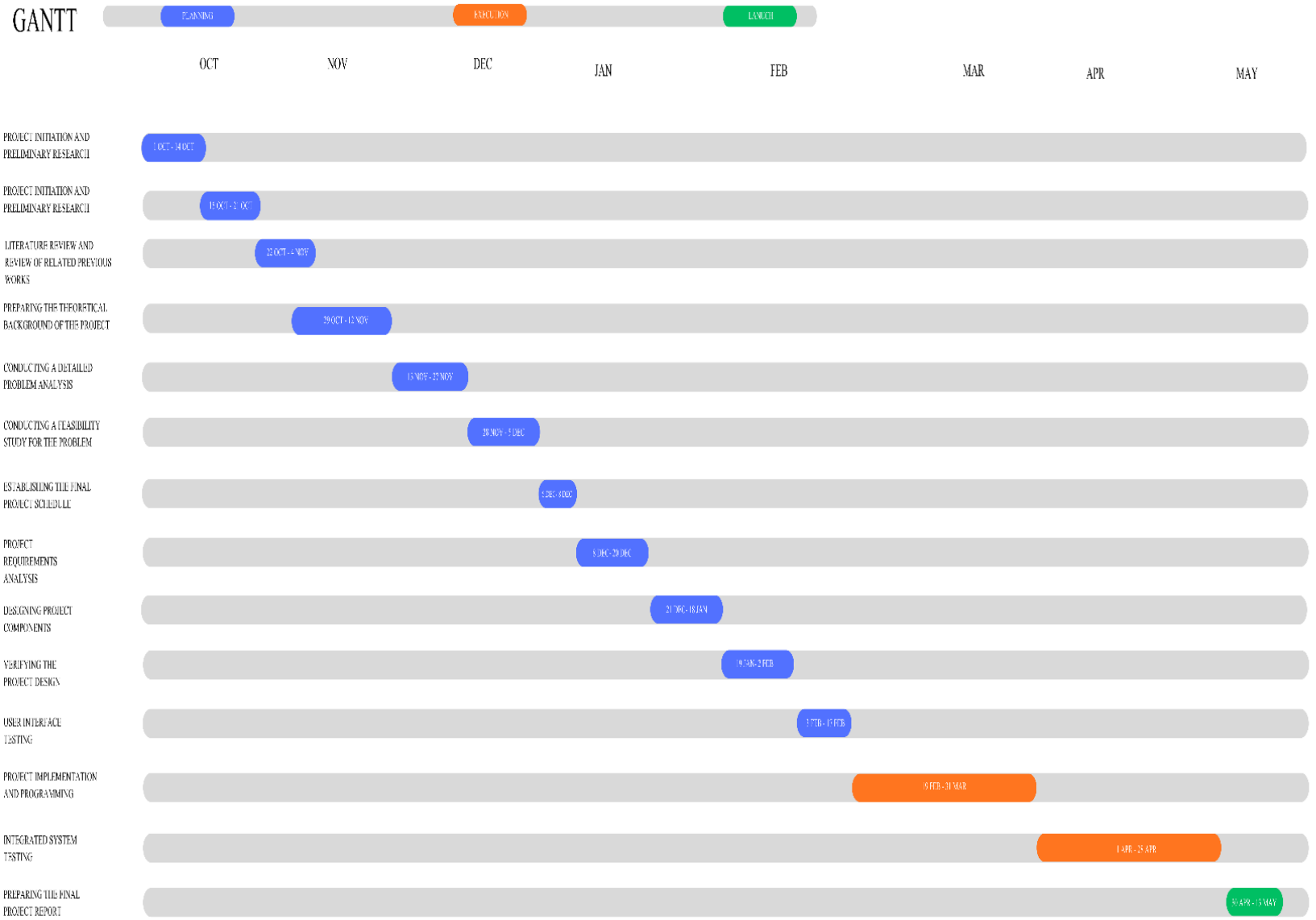
- Research & Development:
- Costs associated with building, training, and testing the AI models and developing the software (app, backend).
- Hardware & Infrastructure:
- The cost of purchasing the physical components: cameras, electronic sensors, and smart gates.
- Operational Costs (Ongoing):
- Server and cloud hosting fees for the IoT platform , backend database , and mobile app.
- Maintenance and improvement of the system

Revenue Stream

- Usage-Based Fees (B2C):
- Drivers pay parking fees via the app; 10–15% is retained as a platform commission, and the rest goes to parking owners.
- B2B Revenue:
- Installation & Setup Fee: One-time fee for hardware/software setup.
- SaaS Subscription: Monthly or annual plans (Basic, Pro, Enterprise) for management tools, analytics, and support.
- Data Insights & Analysis:
- Provide aggregated, anonymized parking insights to facility owners and municipalities for better operations and planning.
- Additional Streams:
- Maintenance & Support Contracts (annual).
- Advertising & Partnerships with nearby businesses.
- Pilot Funding from innovation or startup programs.

3.4 Project Time Scheduling

Activity	Duration
Project Initiation and Preliminary Research	October 1, 2025 – October 14, 2025
Defining Main Objectives and Project Applications	October 15, 2025 – October 21, 2025
Literature Review and Review of Related Previous Works	October 22, 2025 – November 4, 2025
Preparing the Theoretical Background of the Project	October 29, 2025 – November 12, 2025
Conducting a Detailed Problem Analysis	November 13, 2025 – November 27, 2025
Conducting a Feasibility Study for the Problem	November 28, 2025 – December 5, 2025
Establishing the Final Project Schedule	December 6, 2025 – December 8, 2025
Project Requirements Analysis	December 8, 2025 – December 20, 2025
Designing Project Components	December 21, 2025 – January 18, 2026
Verifying the Project Design	January 19, 2026 – February 2, 2026
User Interface Testing	February 3, 2026 – February 17, 2026
Project Implementation and Programming	February 19, 2026 – March 31, 2026
Integrated System Testing	April 1, 2026 – April 28, 2026
Preparing the Final Project Report	April 30, 2026 – May 13, 2026


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JAN
FEB
MAR
APR
MAY
PROJECT INITIATION AND PRELIMINARY RESEARCH
1 OCT - 14 OCT
PROJECT INITIATION AND PRELIMINARY RESEARCH
15 OCT - 21 OCT
LITERATURE REVIEW AND REVIEW OF RELATED PREVIOUS WORKS
22 OCT - 4 NOV
PREPARING THE THEORETICAL BACKGROUND OF THE PROJECT
29 OCT - 12 NOV
CONDUCTING A DETAILED PROBLEM ANALYSIS
15 NOV - 27 NOV
CONDUCTING A FEASIBILITY STUDY FOR THE PROBLEM
28 NOV - 5 DEC
ESTABLISHING THE FINAL PROJECT SCHEDULE
5 DEC - 8 DEC
PROJECT REQUIREMENTS ANALYSIS
8 DEC - 20 DEC
DESIGNING PROJECT COMPONENTS
21 DEC - 18 JAN
VERIFYING THE PROJECT DESIGN
19 JAN - 2 FEB
USER INTERFACE TESTING
3 FEB - 15 FEB
PROJECT IMPLEMENTATION AND PROGRAMMING
16 FEB - 31 MAR
INTEGRATED SYSTEM TESTING
1 APR - 25 APR
PREPARING THE FINAL PROJECT REPORT
30 APR - 15 MAY

3.5 Functional Requirements

This section outlines the core functional requirements of the AI-Powered Smart Parking System. These define what the system must do to provide a fully automated, efficient, and user-friendly parking management solution.

Vehicle Detection and Identification · The system shall automatically detect vehicles entering and exiting the parking area using CCTV cameras and IoT sensors. · The License Plate Recognition (LPR) module shall capture and process vehicle plate images using OpenCV and Tesseract OCR for character recognition. · Each recognized plate number shall be stored securely in the system database, along with time and image records. · In case of recognition failure, the admin shall have the ability to manually verify or enter plate information through the control dashboard.

Parking Slot Monitoring · Every parking slot shall be equipped with an IoT sensor to determine whether the space is occupied or available. · The system shall update parking slot status in real time and display it visually on the dashboard and mobile app. · The parking map interface shall use clear color codes to represent slot states (e..g., green for available, red for occupied, yellow for reserved). · The system shall record the entry and exit times of vehicles to calculate parking duration automatically.

Slot Allocation and Reservation · The system shall automatically assign available slots based on proximity and vehicle size/type. · Users shall be able to search for and reserve slots through the mobile application before arriving. · Once reserved, a slot shall be locked for the user and marked as 'Reserved' to prevent double booking. · The system shall automatically release the slot if the user fails to arrive within the defined time limit (e.g., 15 minutes). · Admins shall have authority to manually override reservations in case of emergencies or system errors.

User Registration and Authentication · New users shall be able to create accounts by registering their personal and vehicle details. · Registered users shall log in securely using their username and password credentials. · The system shall allow users to link multiple vehicles under the same account and update details when needed. · Authentication processes shall ensure data integrity and prevent unauthorized access.

Mobile Application Features · The mobile app shall allow users to: · - View real-time parking availability. · - Reserve and cancel reservations. · - Navigate to the parking facility via integrated GPS navigation. · - Complete digital payments

through secure gateways. · The app shall send push notifications for reservation confirmation, slot expiration, and payment success. · Users shall be able to view their parking history and payment transactions within the app.

Payment and Billing · The system shall automatically calculate parking fees based on duration, vehicle type, and location. · Users shall be able to pay via credit/debit cards, mobile wallets, or other supported online methods. · All financial transactions shall be securely encrypted to ensure user privacy. · Electronic receipts shall be generated and stored in the system database for user and admin reference. · Refunds shall be processed automatically for valid cancellations within the allowed timeframe.

Administrative Control Panel · The admin dashboard shall provide full control and monitoring capabilities, including: · - Viewing real-time parking occupancy. · - Managing user accounts and system access. · - Reviewing transaction logs and reports. · - Monitoring sensor and camera functionality. · Admins shall be able to adjust system parameters such as slot configuration, pricing, and reservation time limits. · Statistical and analytical reports (e.g., daily usage, peak hours, and revenue) shall be generated automatically.

Notification and Alert System · The system shall send real-time notifications and alerts to both users and administrators regarding: · - Successful bookings, payments, and cancellations. · - Slot availability updates. · - Unauthorized vehicle detection or technical faults. · Alerts shall be delivered through in-app notifications, email, or SMS depending on user settings.

System Integration and Communication · All subsystems (IoT sensors, cameras, mobile app, and database) shall be connected to a central control server for synchronized operation. · Data exchange between components shall occur through secure APIs ensuring consistent updates and reliability. · The system shall maintain real-time communication between parking status updates and the user-facing interface.

Reporting and Data Management · The system shall maintain a comprehensive database containing vehicle logs, transactions, and system activities. · Authorized users shall be able to generate and export detailed reports in formats such as PDF or CSV. · The system shall automatically back up important data and remove outdated records according to defined retention policies to ensure efficiency and storage optimization.

3.6 Non-Functional Requirements

Non-Functional Requirements for “AI-Powered Smart Parking System” Performance The system will be designed to provide fast and efficient performance across all main components: · License Plate Recognition (LPR): The model, built using OpenCV and Tesseract OCR, is expected to detect and recognize vehicle license plates in around 1 second per image under normal lighting conditions. · Sensor and Camera Updates: IoT sensors and cameras will send data to the main server and update parking status within 2 seconds after detecting a vehicle entering or leaving a spot. · Mobile App Performance: The mobile application will handle user requests (like searching for available spots or booking) with a response time of 2–3 seconds on average. · System Load: The system can manage data from about 10–15 cameras and serve up to 100 active users at the same time without performance issues. If traffic increases, the system will use basic queue management techniques to process requests in order without crashes. Security Security is a key part of the project to ensure that both user and vehicle data remain safe. · Data Encryption: Sensitive data such as license plate numbers and user information will be encrypted using AES (Advanced Encryption Standard) before being stored in the database. Communication between the mobile app, IoT devices, and the server will be protected using HTTPS (SSL). · Authentication: The system will include secure login with username and password, and will also implement two-factor authentication (2FA) for user login to add an extra layer of protection. · Account Protection: If there are multiple failed login attempts (e.g., 5 tries in 10 minutes), the account will be temporarily locked to prevent unauthorized access. The admin will be able to unlock accounts manually from the control panel Reliability The system aims for about 98–99% accuracy in recognizing license plates in normal conditions. · Handling Low Accuracy Situations: If the recognition accuracy drops below 95% because of environmental factors (like rain, poor lighting, or dirty plates), the system will switch to a manual review mode, where the admin can verify the detected plates through the dashboard. · Error and Alert Handling: When the system fails to detect a plate or a sensor stops sending data, an alert message will appear for the admin, and the issue will be logged for review. · Backup Mode: If the internet connection is lost, local functions (like opening gates or updating sensors) will keep running locally until the connection is restored. Usability

- The user interface (UI) of both the mobile app and the web platform should be simple, intuitive, and easy to navigate for ordinary users.
- Clear colors and icons should be used to indicate parking status (e.g., occupied / available).

- The system should support at least two languages (Arabic and English).

Scalability

- The system should be designed to scale easily and support multiple parking facilities across different cities.
- The database and server architecture should accommodate an increasing number of users and vehicles without major reconfiguration.

Maintainability

- The source code should be well-documented and modularized to simplify maintenance and future updates.
- Software updates should be deployable without service interruption.

Availability

- The system must be available 24/7 to users and administrators.
- In case of downtime, the system should automatically notify the technical support team.

Compatibility

- The mobile application must be compatible with Android and iOS platforms.
- The administrative dashboard should function properly on all modern web browsers (Chrome, Edge, Firefox).

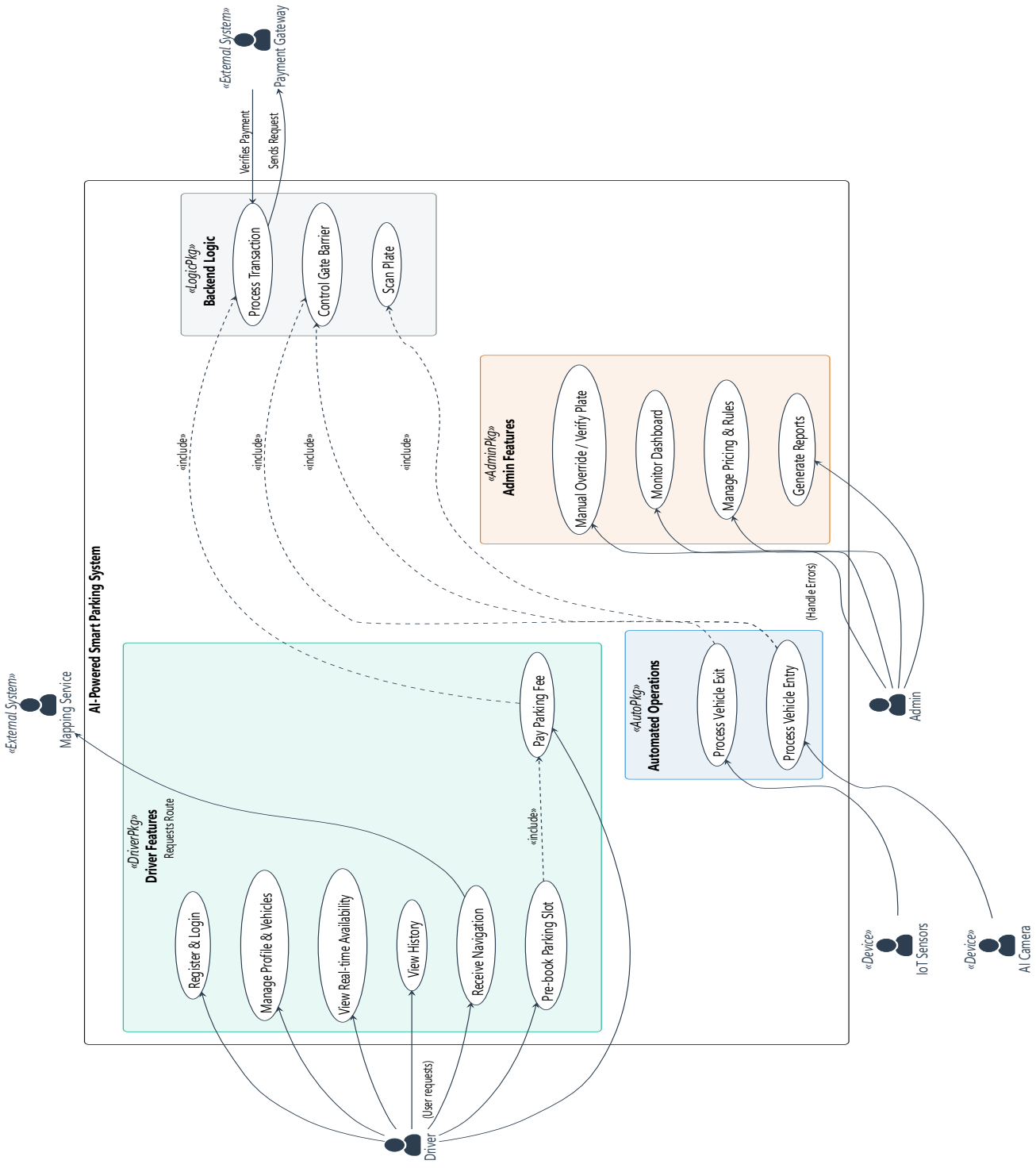
Backup and Recovery

- A daily backup of the database must be maintained.
- In the event of system failure, full recovery should be achievable within 10 minutes.

Privacy

- Images and license plate data should be used solely for administrative and security purposes.
- No personal data should be stored or shared without the user's explicit consent.

3.7 Use Case Diagram



3.8 Use case scenario

1. Driver Features (App Scenarios)

UC-01: Register & Login

Field	Description
Primary Actor	Driver
Pre-conditions	User has installed the ESPS mobile application.
Main Flow	1. Driver opens the app and selects “Register”. 2. Driver enters Name, Phone Number, and Password. 3. System sends an OTP (One-Time Password) for verification. 4. Driver enters OTP. 5. System validates and creates the account. 6. Driver logs in successfully.
Post-conditions	Driver is authenticated and redirected to the Home Screen.

UC-02: Manage Profile & Vehicles

Field	Description
Primary Actor	Driver
Pre-conditions	Driver is logged in.

Main Flow	1. Driver navigates to “My Vehicles”. 2. Driver clicks “Add Vehicle”. 3. Driver enters License Plate Number and Vehicle Model. 4. Driver uploads a photo of the license plate (optional for validation). 5. System validates the plate format. 6. System saves the vehicle data to the profile.
Post-conditions	Vehicle is linked to the user account for automatic recognition at the gate.

UC-03: View Real-time Availability

Added to match Diagram.

Field	**Description**
Primary Actor	Driver
Pre-conditions	Driver is logged in.
Main Flow	1. Driver opens the “Home” or “Map” view. 2. System queries the database for current occupancy status. 3. System displays available spots count and specific zone status (e.g., Zone A: 5 spots left). 4. Driver views the heatmap or list of free slots.
Post-conditions	Driver is informed of parking availability before arriving.

UC-04: View History

Added to match Diagram.

Field	Description
Primary Actor	Driver
Pre-conditions	Driver is logged in and has past parking activity.
Main Flow	1. Driver navigates to the "History" tab. 2. System retrieves past reservations and completed parking sessions. 3. Driver selects a specific entry to view details (Date, Duration, Cost). 4. System displays the digital receipt.

UC-05: Pre-book Parking Slot

Field	Description
Primary Actor	Driver
Included Use Case	**Pay Parking Fee (UC-06)**
Pre-conditions	Driver is logged in and has a registered vehicle.
Main Flow	1. Driver selects a date and time for parking. 2. System checks database availability. 3. System identifies an available zone internally. 4. System displays the estimated fee and prompts for payment. 5. **System includes Use Case: Pay Parking Fee.** (The booking cannot proceed until payment is successful). 6. Upon successful payment, System confirms booking. 7. System generates a Reservation ID/QR Code.
Alternate Flow	**A1: Fully Booked:** System notifies driver that no slots are available. **A2: Payment Cancelled:** If driver cancels payment, the booking is aborted.
Post-conditions	A slot is reserved, and the transaction is recorded.

UC-06: Pay Parking Fee

Field	Description
Primary Actor	Driver, Payment Gateway (External)
Pre-conditions	Driver has selected a slot to book OR is at the exit gate trying to leave.
Main Flow	1. Driver selects "Pay Now" (or is prompted by Pre-book flow). 2. System calculates total cost. 3. Driver confirms payment method. 4. System sends request to Payment Gateway. 5. **Payment Gateway** verifies and approves transaction. 6. System updates status to "Paid".
Alternate Flow	**A1: Payment Declined:** Gateway returns error; App prompts user to try another card.
Post-conditions	Revenue recorded; Booking confirmed or Exit authorized.

UC-07: Receive Navigation

Field	Description
Primary Actor	Driver
Secondary Actor	**External Mapping Service**
Pre-conditions	A slot has been allocated.
Main Flow	1. Driver clicks "Navigate to Slot" on the app. 2. System retrieves the coordinates of the assigned slot. 3. System sends coordinates to the **External Mapping Service**. 4. External Mapping Service generates the route. 5. App displays the path from current location to the specific slot.

Field	Description
Primary Actor	AI Camera (Device)
Included Logic	Scan Plate, Control Gate Barrier
Pre-conditions	Vehicle arrives at the entry gate.
Main Flow	1. AI Camera detects motion and triggers the system. 2. System performs **Scan Plate** (Computer Vision) to identify the license number. 3. System checks for existing reservation or availability. 4. System determines slot allocation. 5. System sends slot info to Driver's App. 6. System performs internal safety check. 7. System performs **Control Gate Barrier** (Open). 8. Vehicle enters.
Post-conditions	Gate closes, timer starts, session created.

UC-09: Process Vehicle Exit

Field	Description
Primary Actor	IoT Sensors (Device)
Included Logic	Control Gate Barrier
Pre-conditions	Vehicle arrives at the exit gate.
Main Flow	1. IoT Sensor detects vehicle at exit. 2. System checks payment status for the detected plate. 3. **If Paid:** System proceeds. 4. System performs internal safety check (Ultrasonic sensors). 5. System performs **Control Gate Barrier** (Open). 6. Vehicle leaves.
Alternate Flow	**A1: Not Paid:** Barrier remains closed; Screen prompts driver to pay via app.
Post-conditions	Parking session closed; Slot marked as "Available".

3. Admin Features (Management)

UC-10: Monitor Dashboard

Field	Description
Primary Actor	Admin
Main Flow	1. Admin logs into the Web Dashboard. / 2. System pulls real-time data from database. 3. Dashboard displays: Occupancy Rate, Revenue, Active Alerts. 4. Admin filters view by Zone.

UC-11: Manage Pricing & Rules

Field	Description
Primary Actor	Admin
Main Flow	1. Admin navigates to "Settings". 2. Admin modifies the hourly rate. 3. Admin updates logic rules (e.g., peak hour timings). 4. System saves configuration.

UC-12: Generate Reports

Field	Description
Primary Actor	Admin
Main Flow	1. Admin selects "Reports" section. 2. Admin chooses date range and report type. 3. System generates PDF/Excel file. 4. Admin downloads the report.

UC-13: Manual Override / Verify Plate

Field	Description
Primary Actor	Admin
Pre-conditions	System failed to recognize a plate OR a driver claims a booking error.

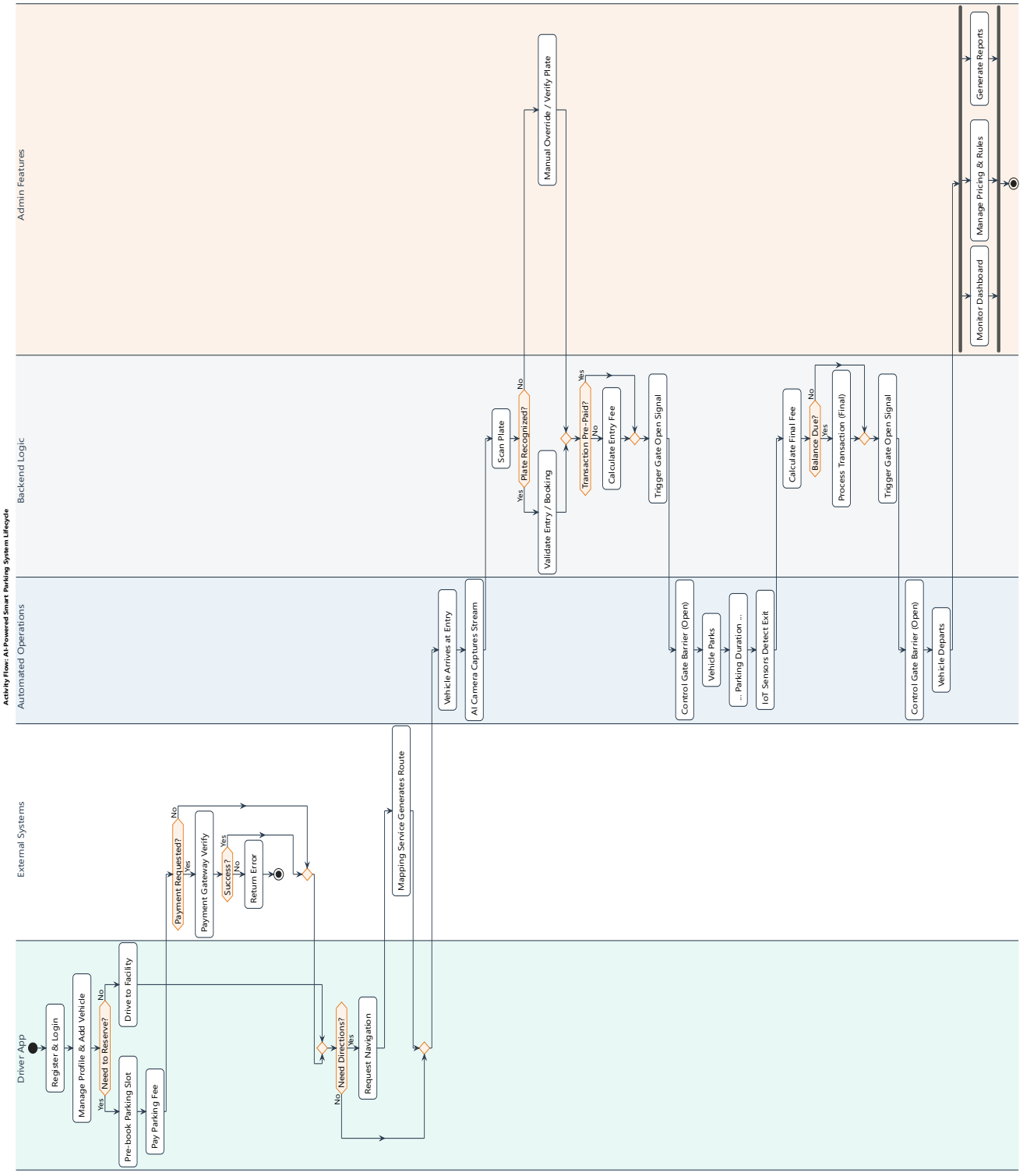
Main Flow	1. System alerts Admin of an "Unidentified Vehicle" or Driver calls for help. 2. Admin views the live camera feed/snapshot on the dashboard. 3. Admin manually enters or corrects the License Plate Number. 4. System validates the manual entry against reservations/database. 5. Admin issues "Force Open Gate" command if verified. 6. System logs the manual override action for security auditing.
Post-conditions	Gate opens, and the vehicle is processed manually.

4. Backend Internal Logic

Sub-routines triggered by other Use Cases.

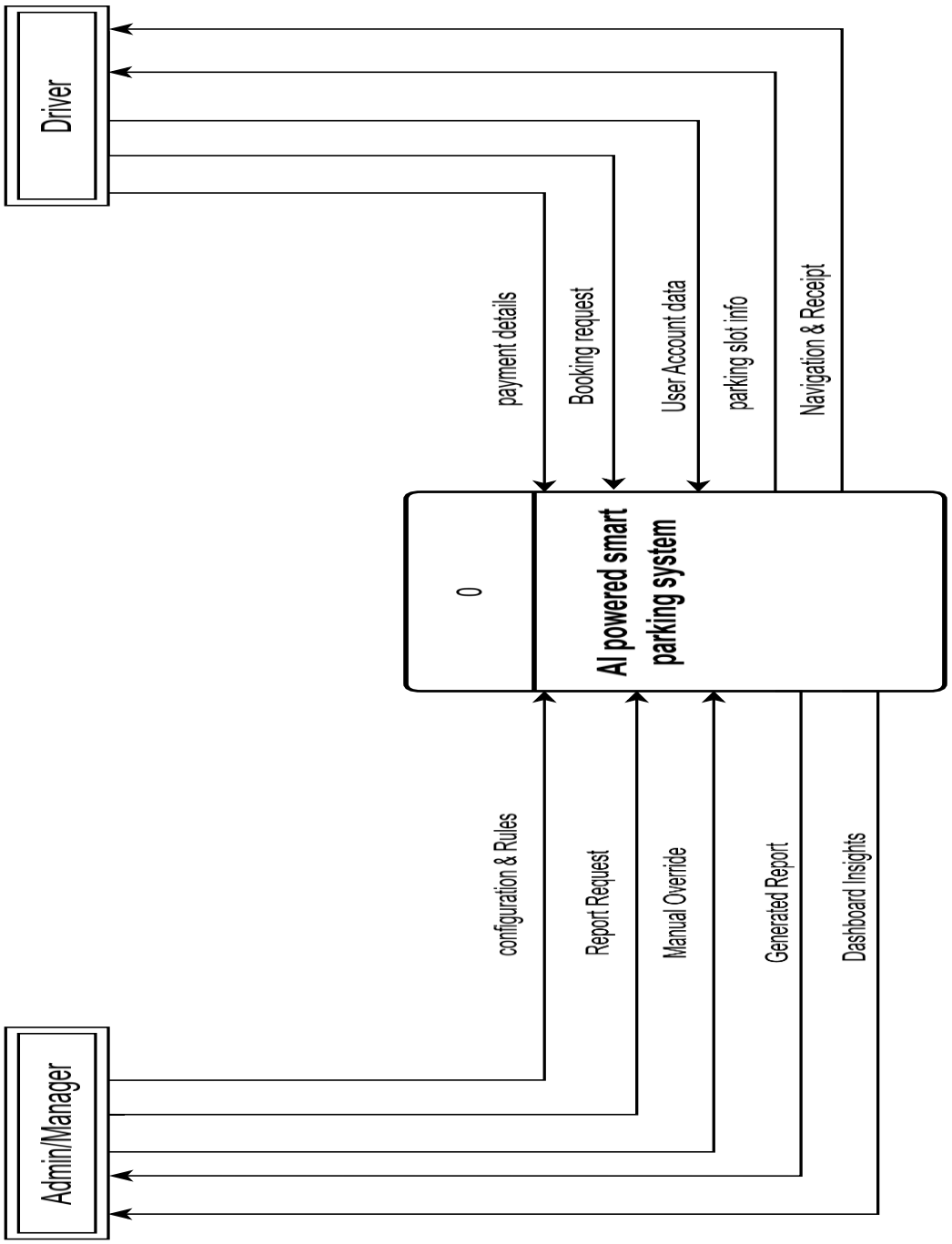
1. **Scan Plate & Count Passengers:** The AI model processing the video frame to output text (Plate) and integer (Passenger Count).
2. **Control Gate Barrier:** The signal sent to the Arduino/Servo to rotate 90 degrees.
3. **Process Transaction:** The secure API call to the Banking System.

3.9 Activity Diagram

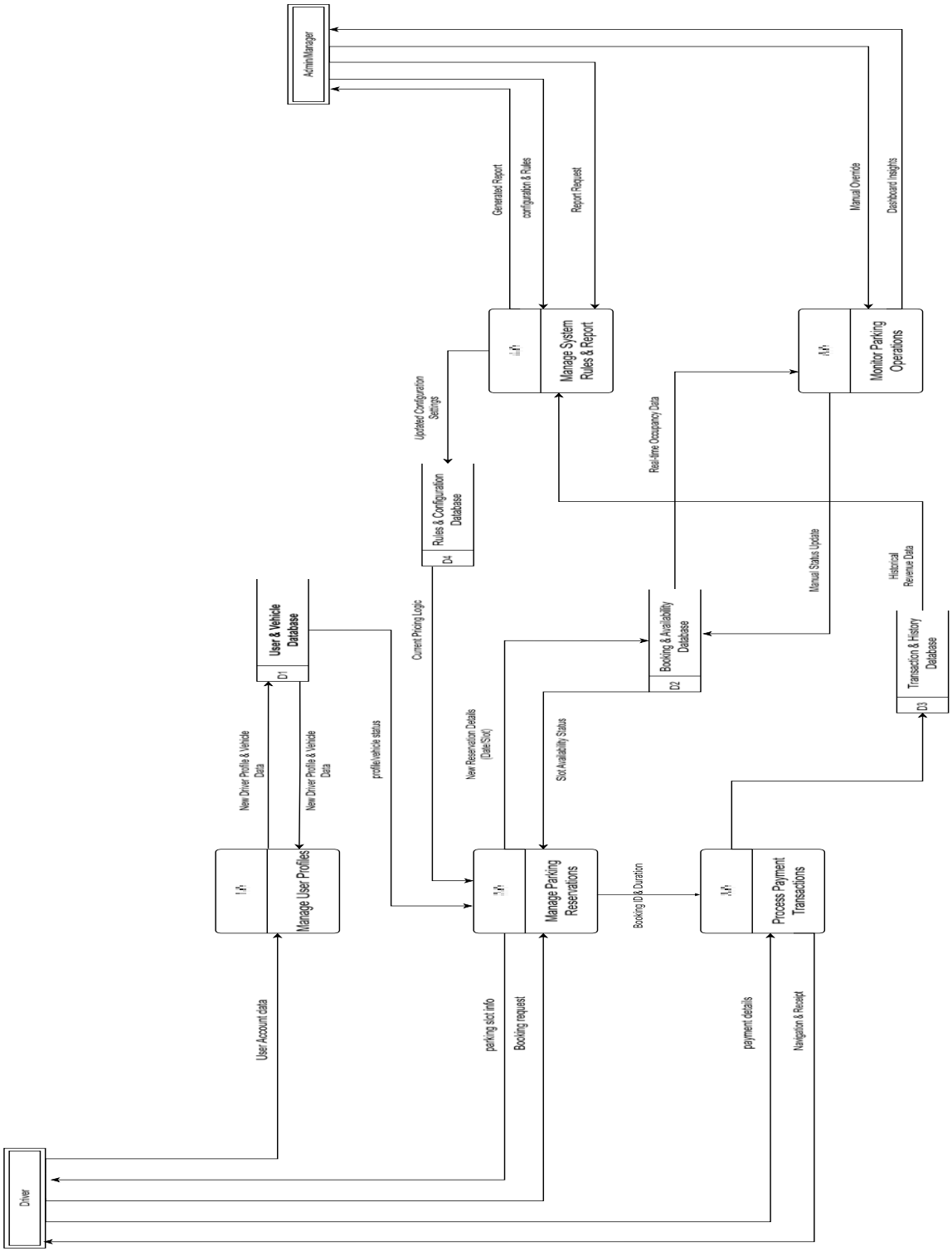


3.10 Data Flow Diagram

3.10.1 Context diagram

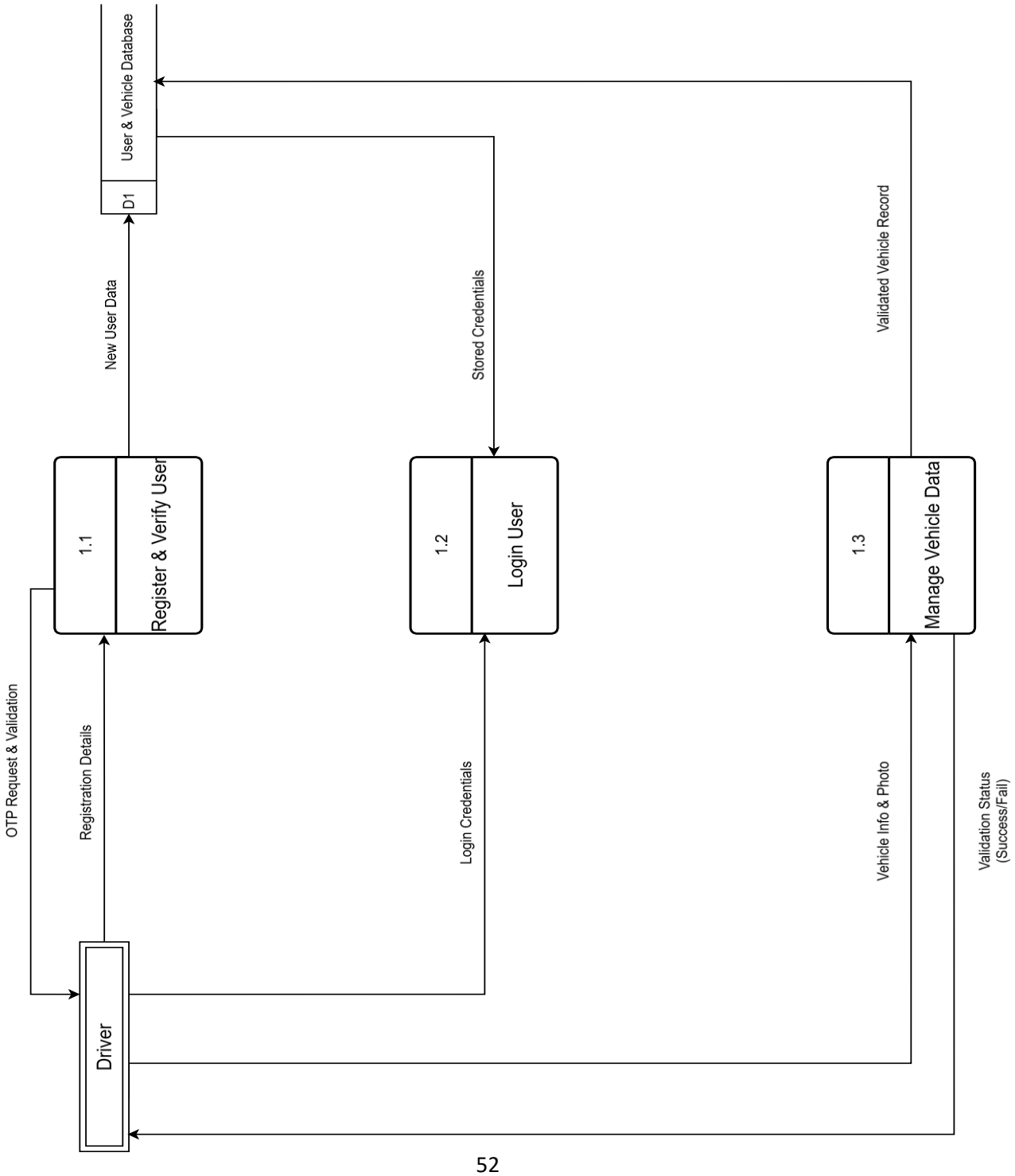


3.10.2 Level 0 diagram

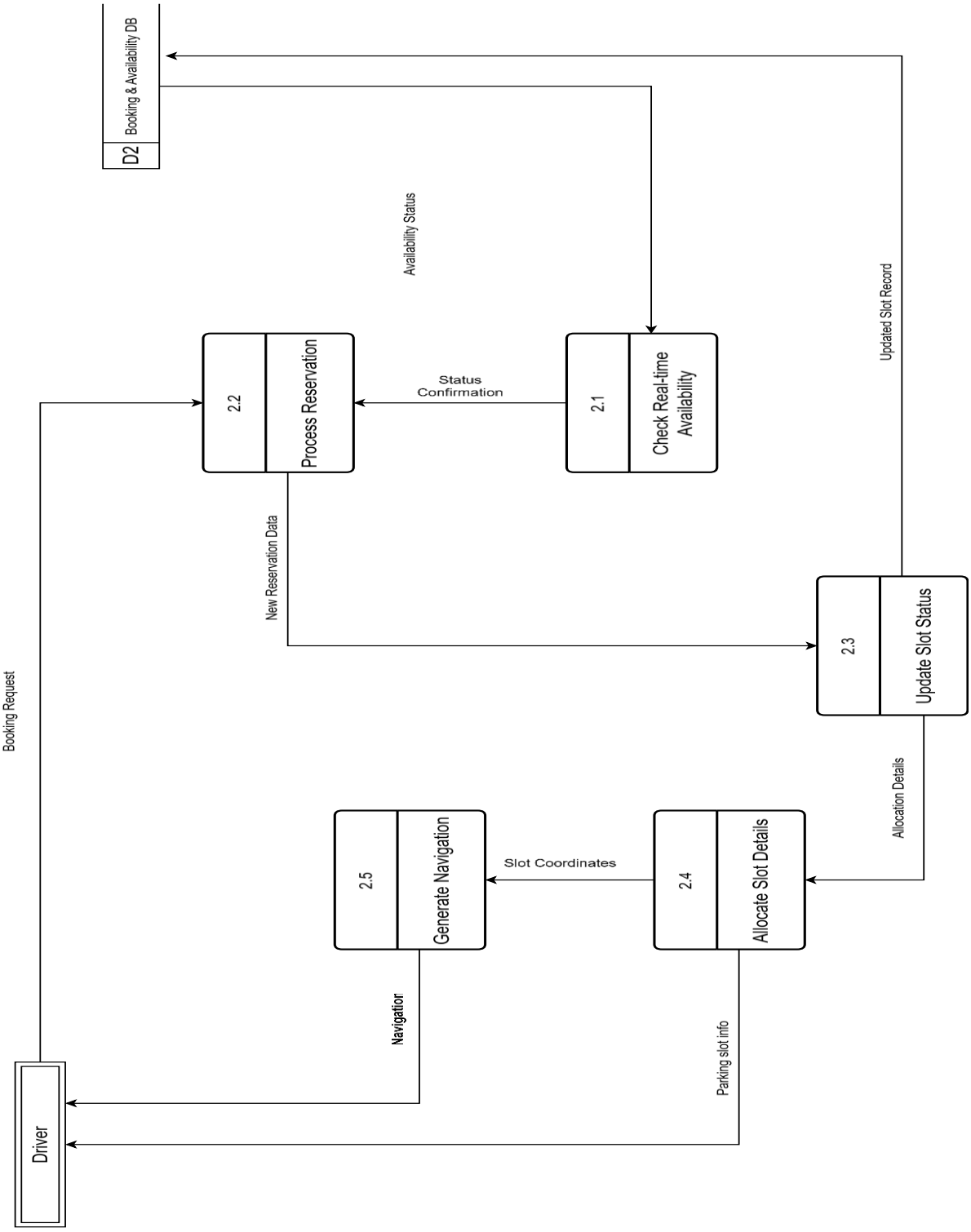


3.10.3 Level 1 Diagram

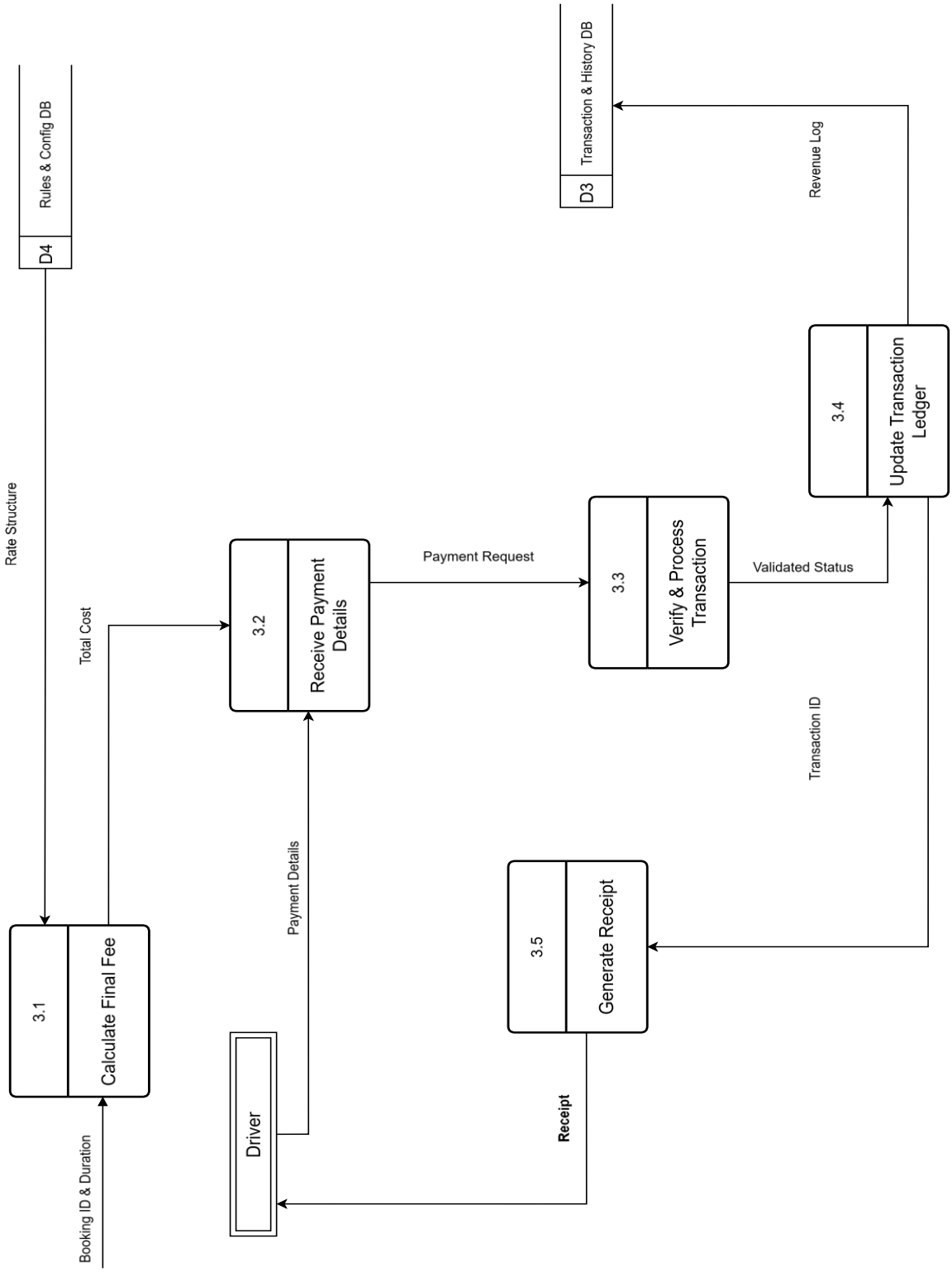
3.10.3.1 Process 1



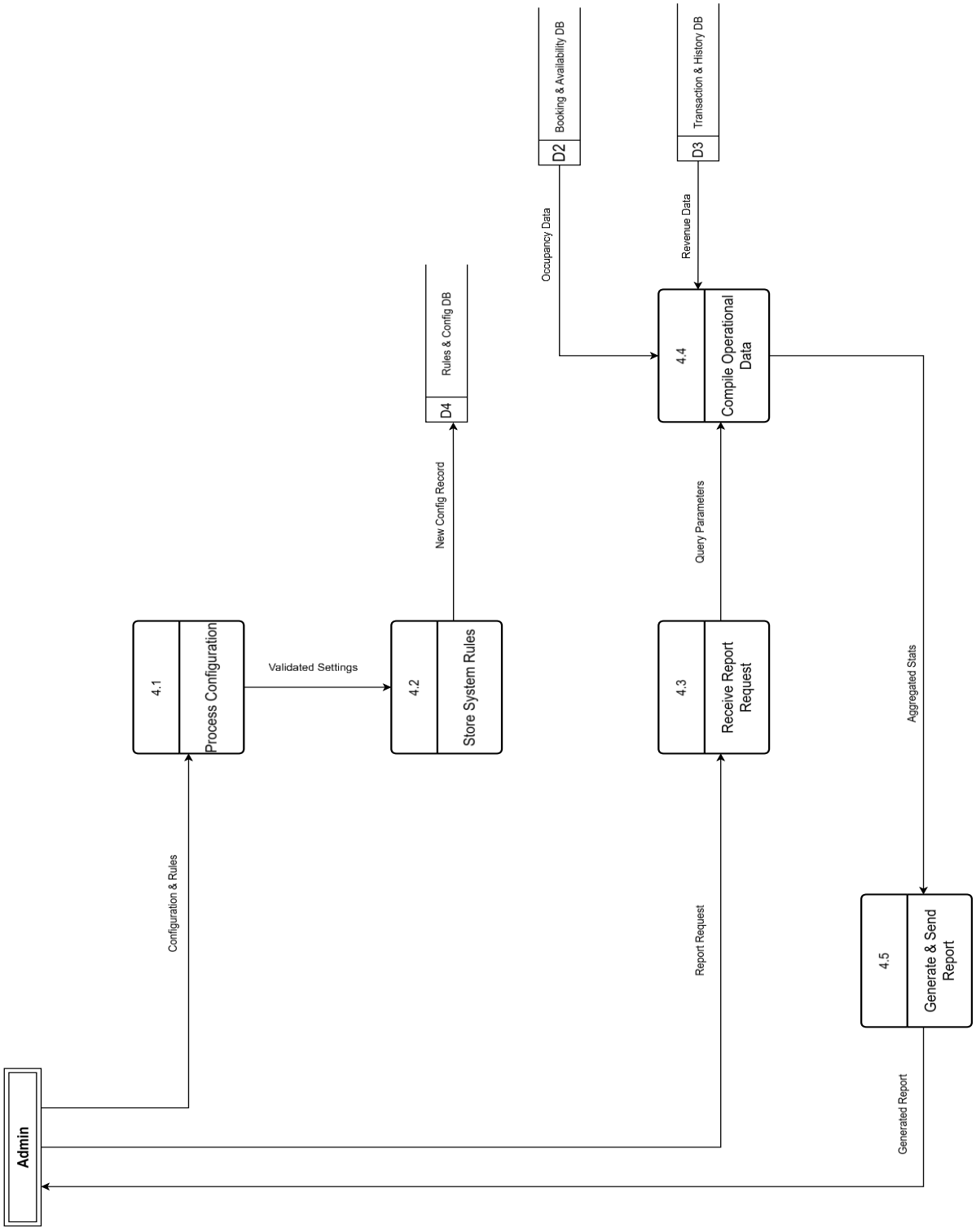
3.10.3.2 Process 2



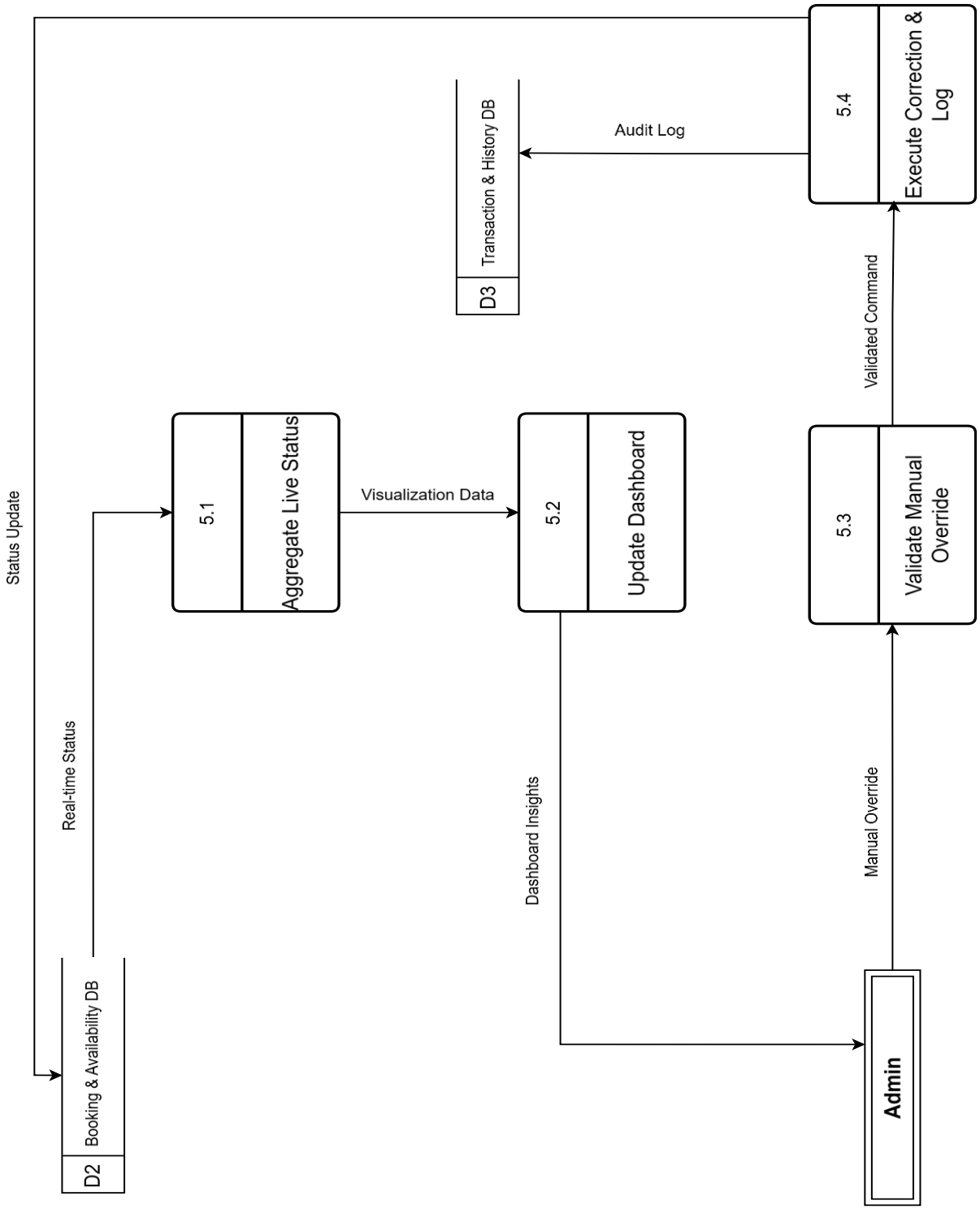
3.10.3.3 Process 3



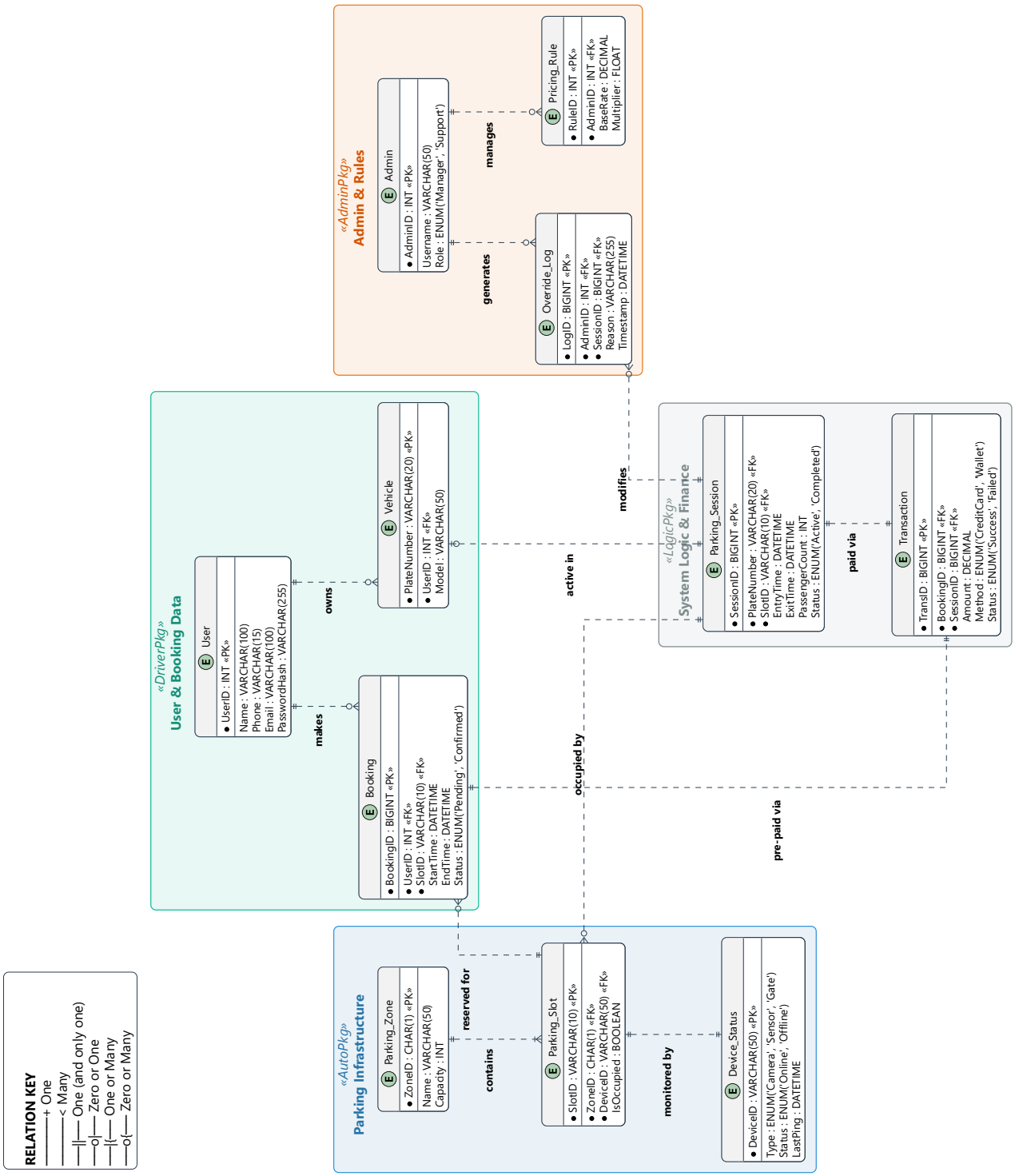
3.10.3.4 Process 4



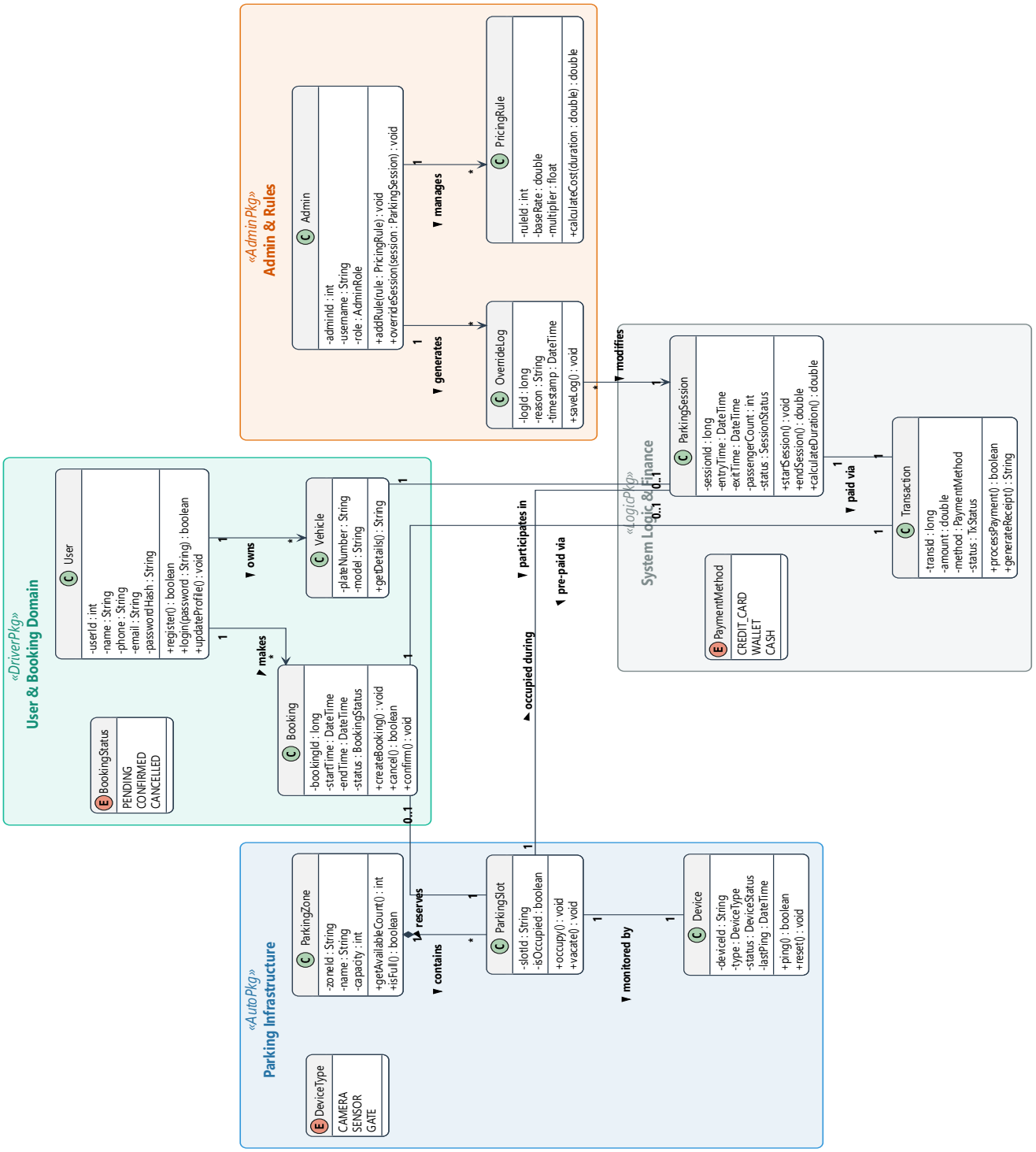
3.10.3.5 Process 5



3.11 Entity Relationship Diagram

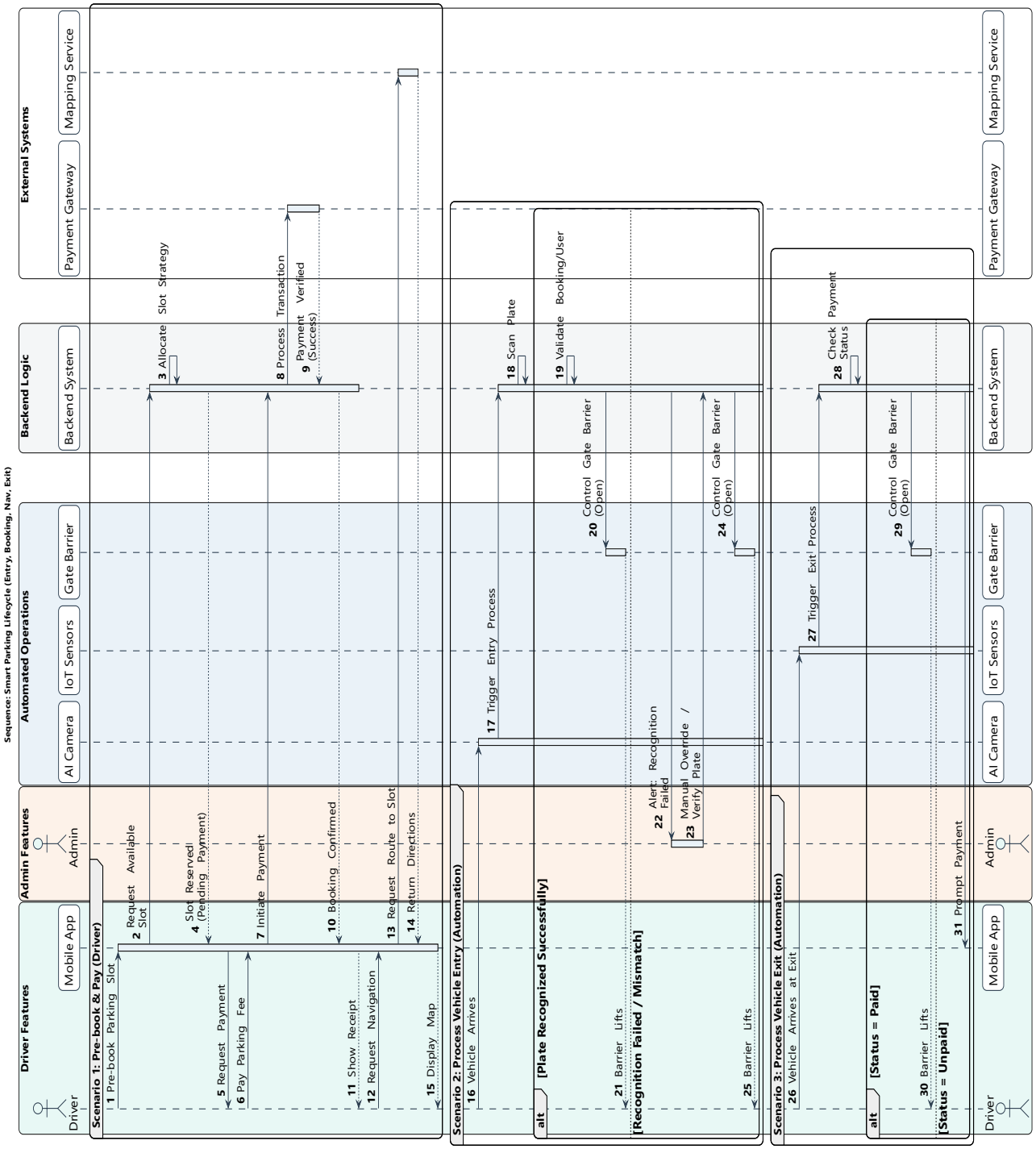


3.12 Class diagram

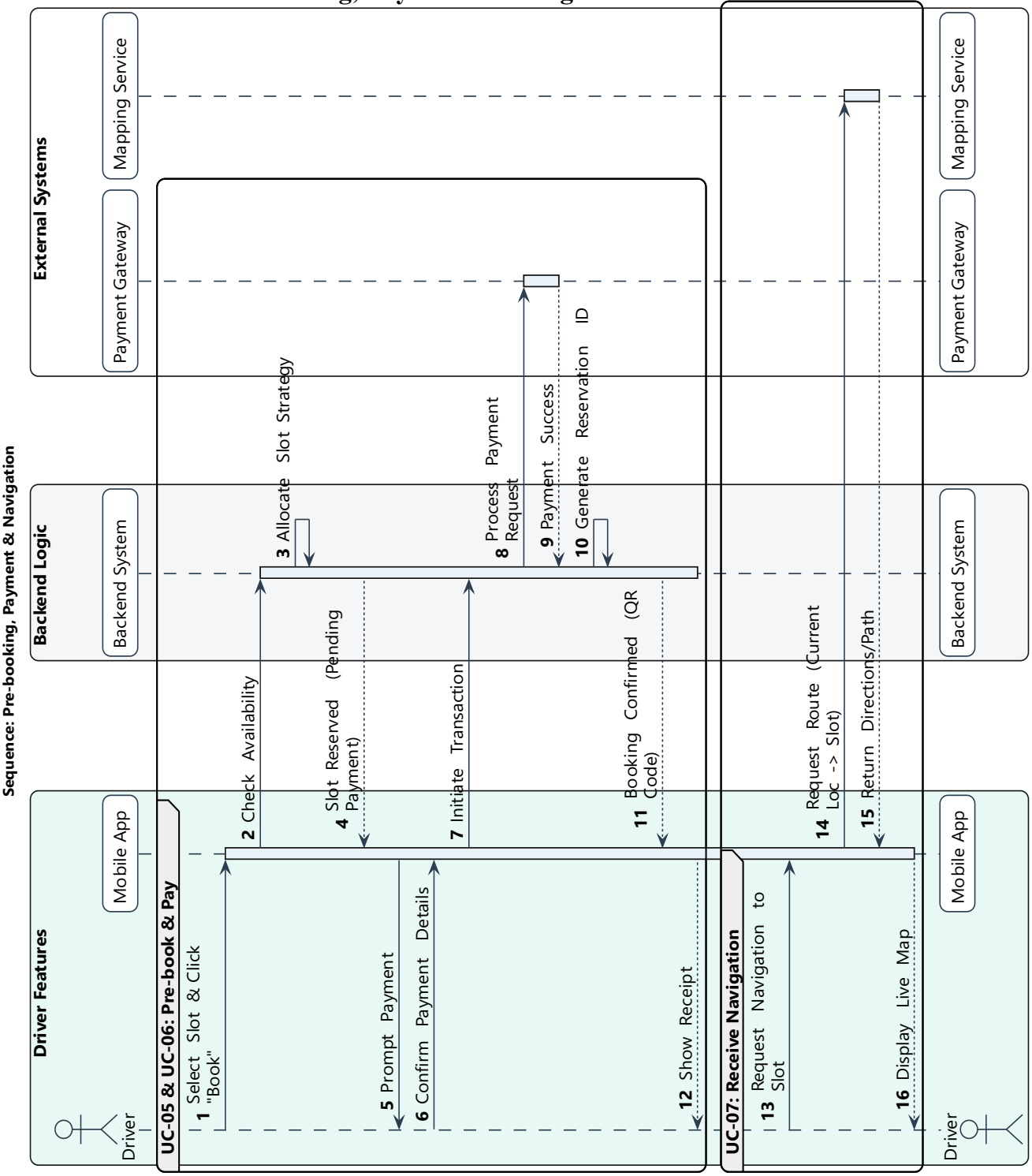


3.13 Sequence diagram

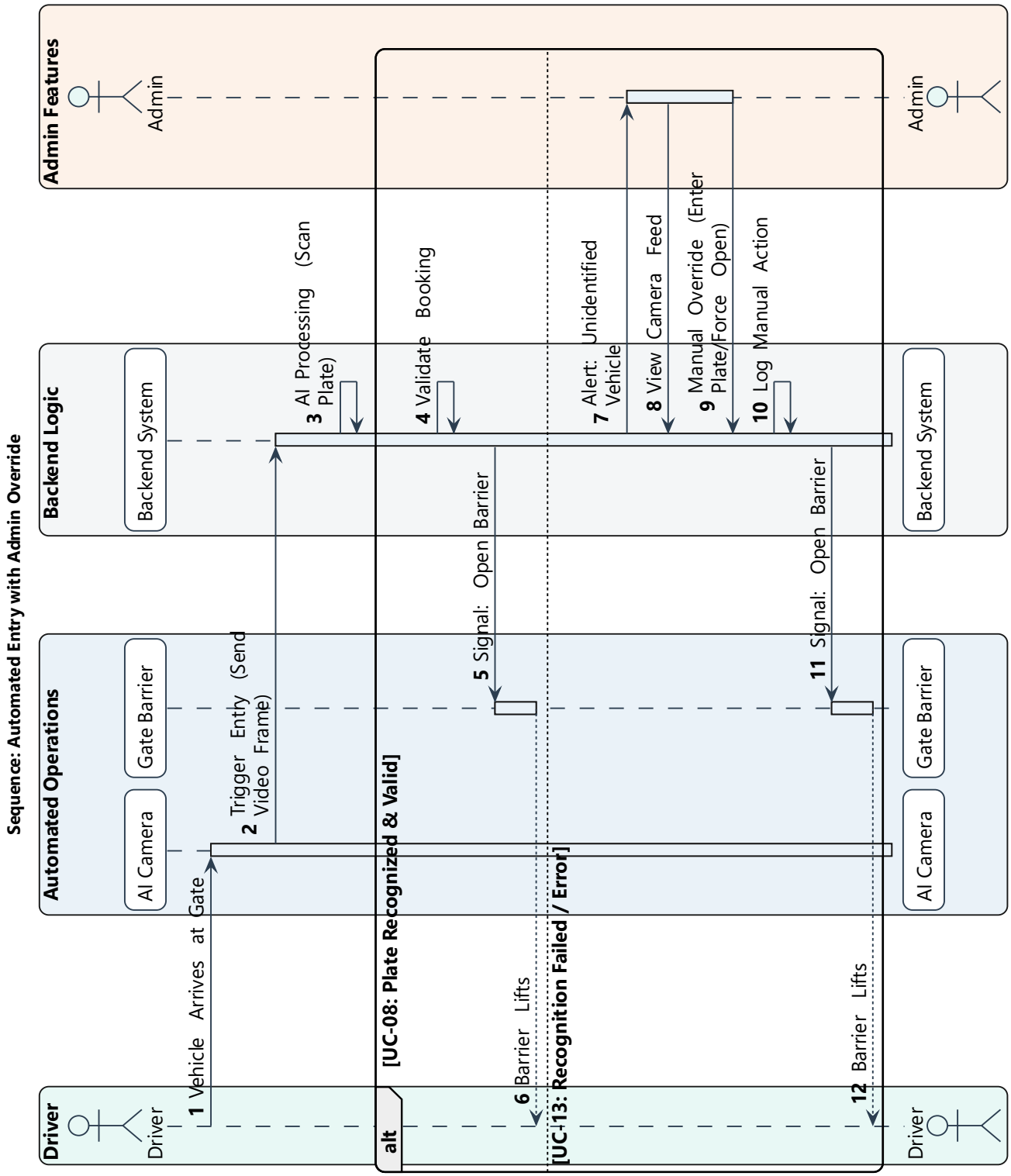
3.13.1 Full system sequence diagram



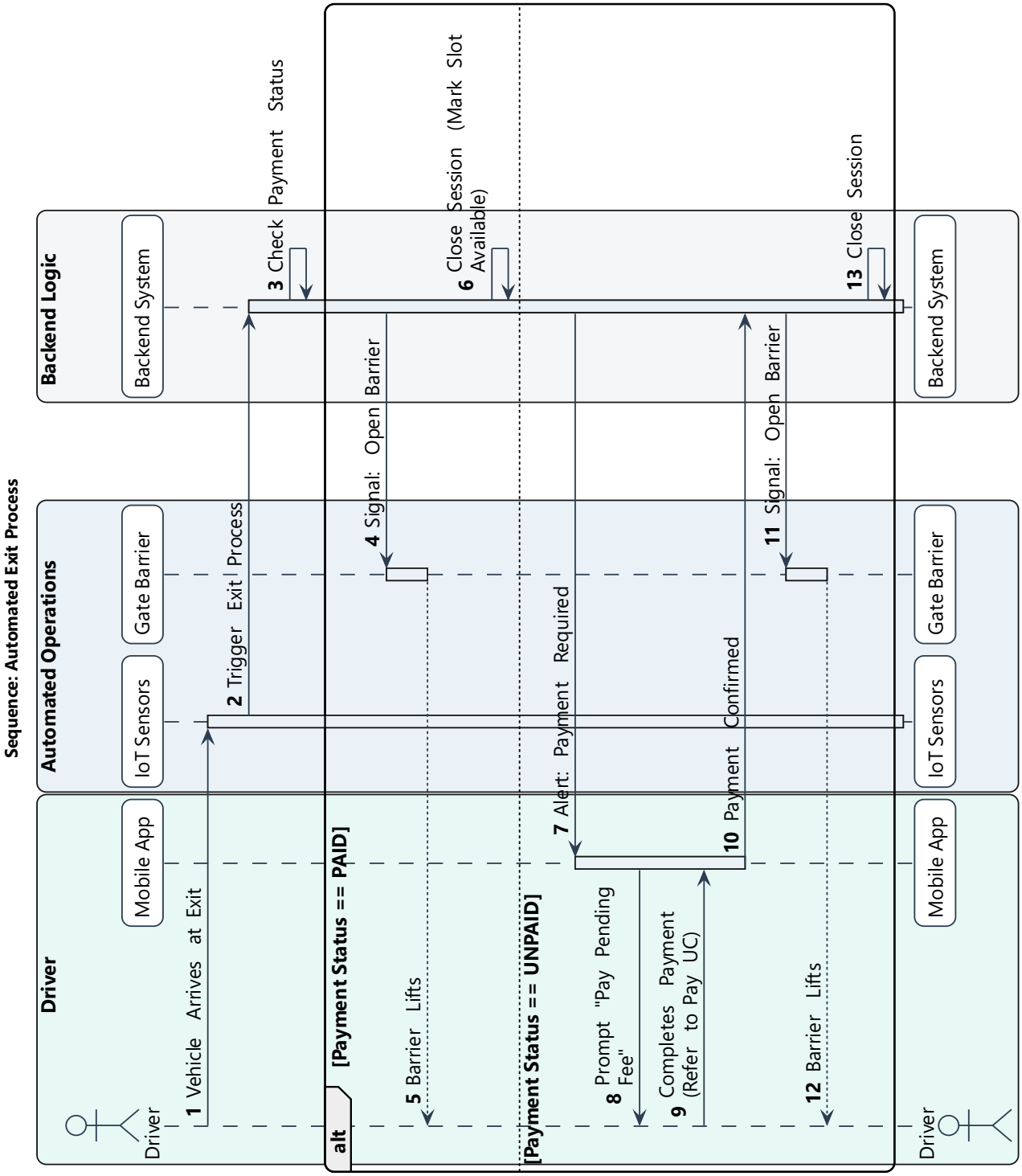
3.13.2 Pre-booking, Payment & Navigation



3.13.3 Vehicle Entry & Admin Intervention



3.13.4 Vehicle Exit Process



Chapter 4 PROJECT DESIGN

4.1. Design System & Aesthetics

Visual Theme: "Night Mode" aesthetic utilizing a high-contrast Dark Grey/Black background (#1E1E1E) with Neon Lime Green (#CCFF00) accents for primary actions and highlights.

Typography: Modern sans-serif font family, ensuring readability and a futuristic tech feel.

Navigation: Standard Bottom Navigation Bar (Home, Saved, Activity, Profile) for consistent user routing.

4.2. User Flow & Key Modules

A. Onboarding & Authentication

Splash Screen: Displays the application logo ("Rakni") with a minimal centralized branding approach.

Walkthrough: A carousel introduction highlighting key value propositions:

Real-time Slot Availability.

EV Charging Station locators.

Secure Login:

User-friendly "Login with Phone" interface.

OTP Verification: 4-digit security code entry with success feedback modal.

B. Discovery & Booking Core

Home Dashboard (Map View):

Google Maps integration showing current user location and nearby parking lots (e.g., "Cairo Festival City Parking").

Quick-info cards displaying distance, pricing, and available spots.

Slot Selection Interface:

Visual Garage Map: A grid representation of the parking floor (A-012, A-013) indicating "Occupied" (Grey) vs. "Available" (Green) slots.

Floor Navigation: Toggles for switching between garage levels.

C. "Smart" Features (AI & IoT Integration)

License Plate Recognition:

"Scan Plate" Interface: A camera viewfinder overlay designed for Optical Character Recognition (OCR) to automatically detect user vehicles or validate entry.

AR/Visual Guidance:

Indoor Navigation: Directional arrows overlay (Augmented Reality style) guiding the driver specifically to their booked slot (e.g., "Turn Right for Parking Spot").

D. Session Management & Payment

Parking Timer:

A circular progress ring displaying "Time Remaining" (e.g., 02:01:20) to prevent overstaying.

"Extend Time" functionality for remote session management.

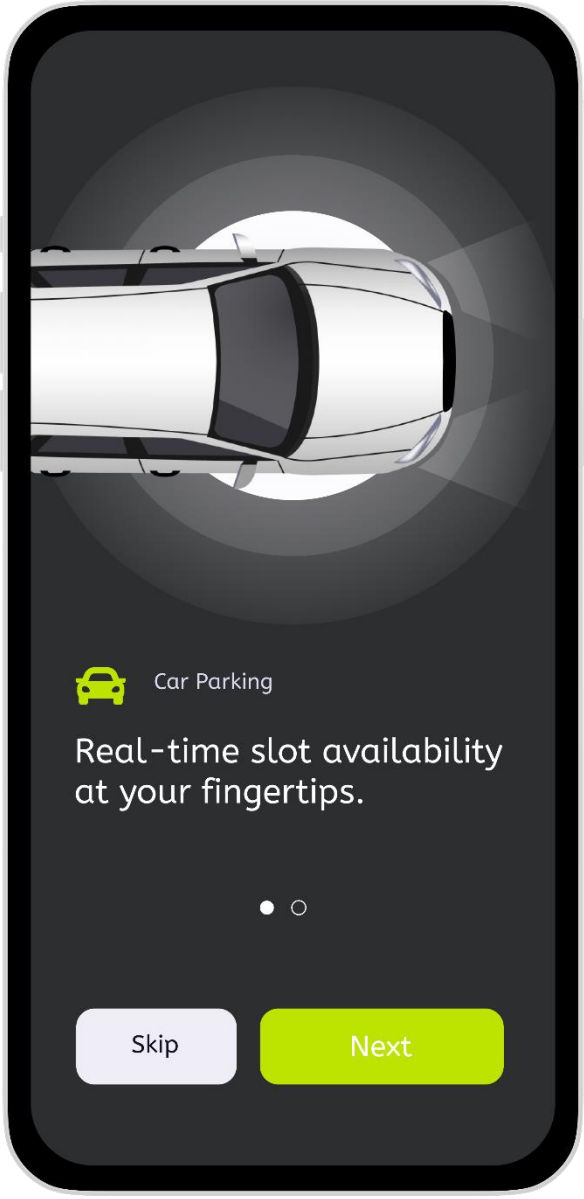
Checkout & E-Ticket:

Payment Gateway: Support for multiple methods (Saved Cards, Credit/Debit).

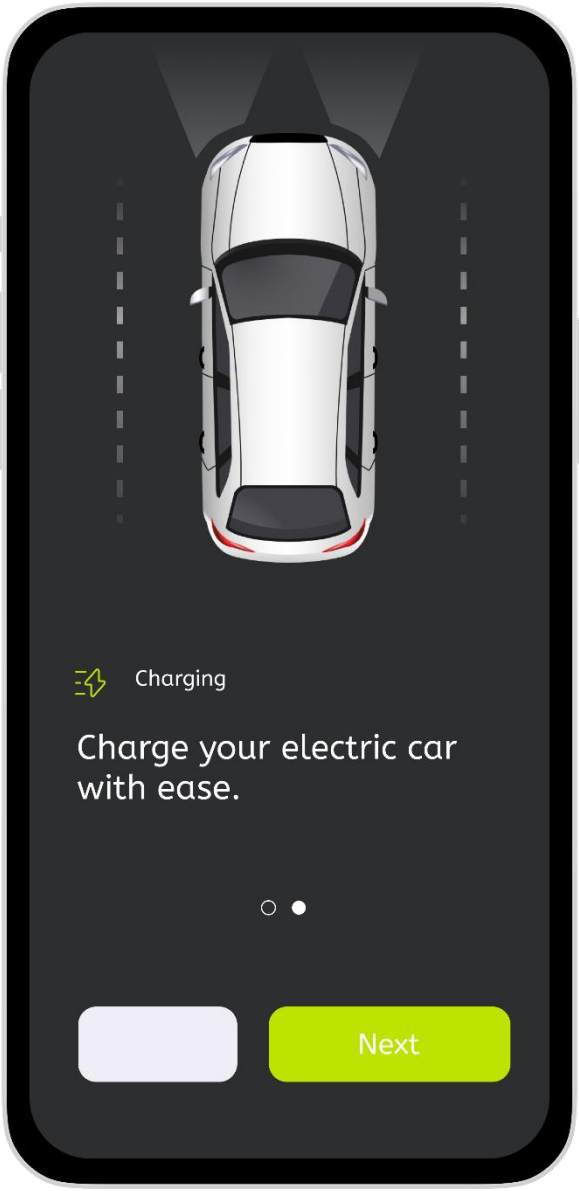
Digital Ticket: A summary screen showing the QR code, User Profile, Booking ID, and total cost for gate validation.



Splash Screen



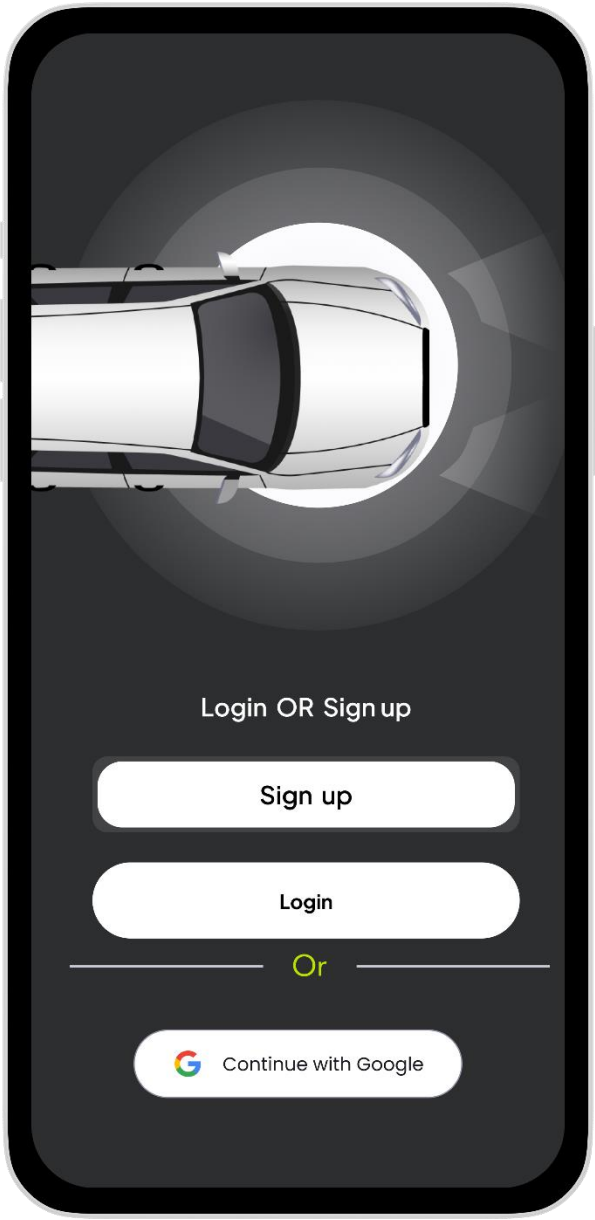
Onboarding



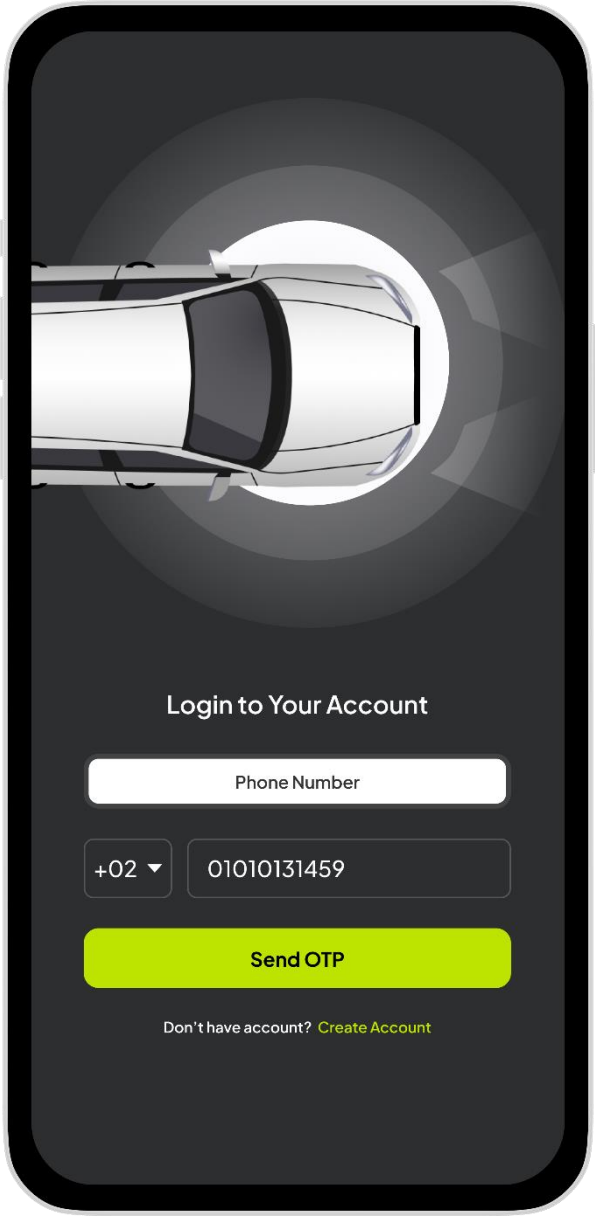
Onboarding Screen



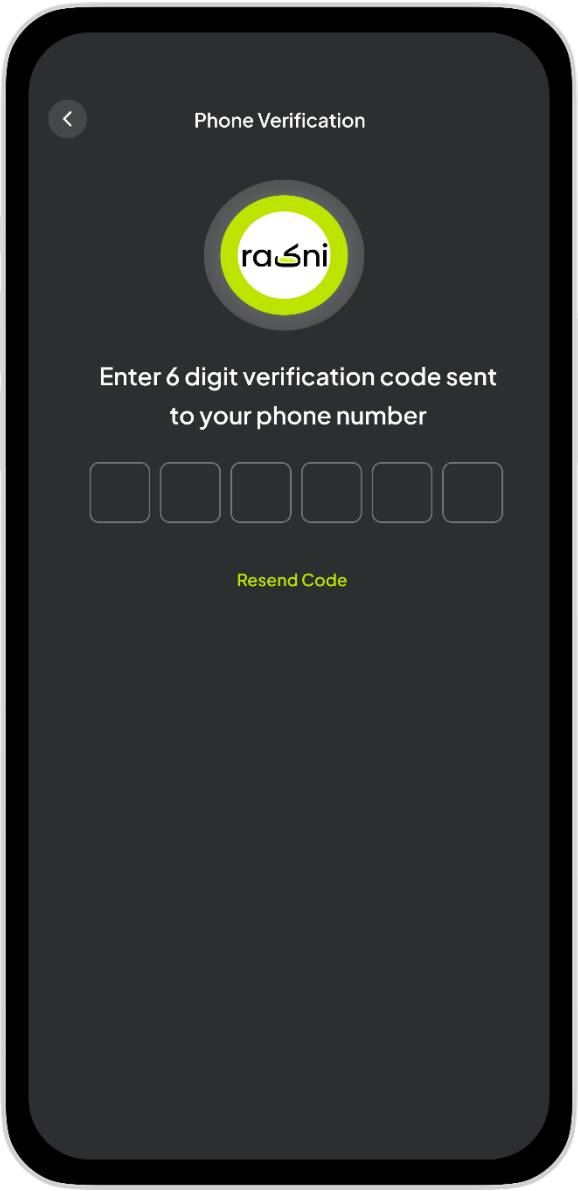
Start Screen

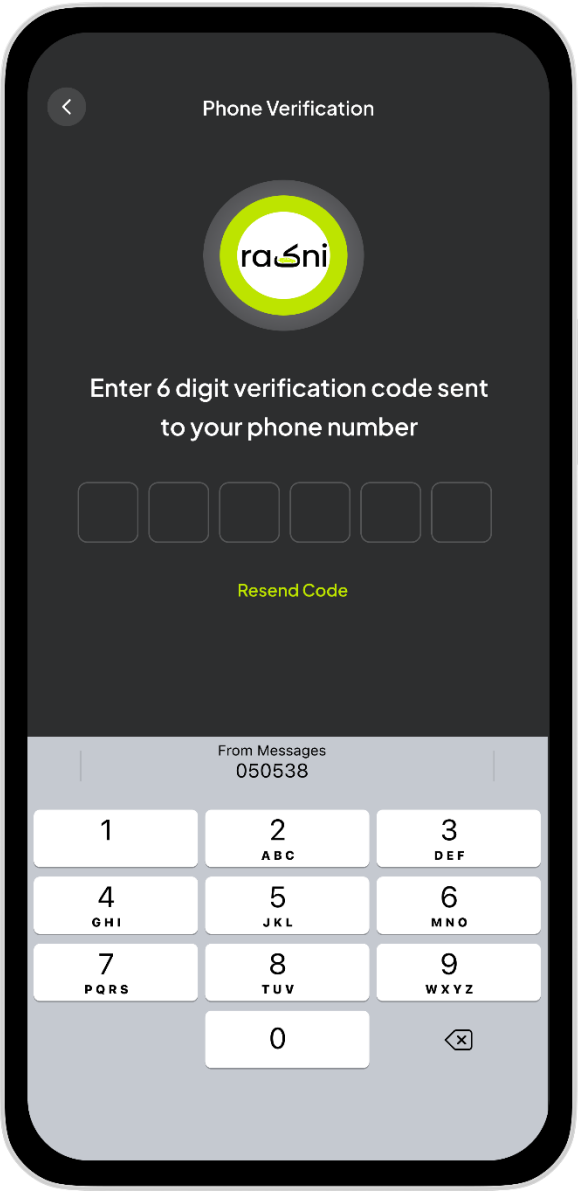


Login Screen

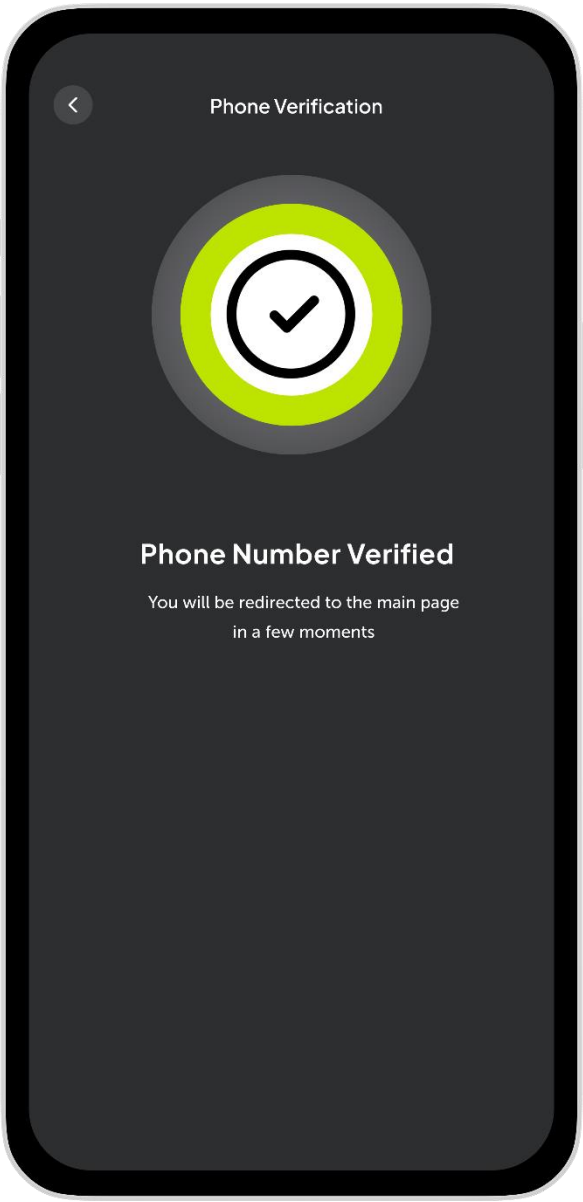


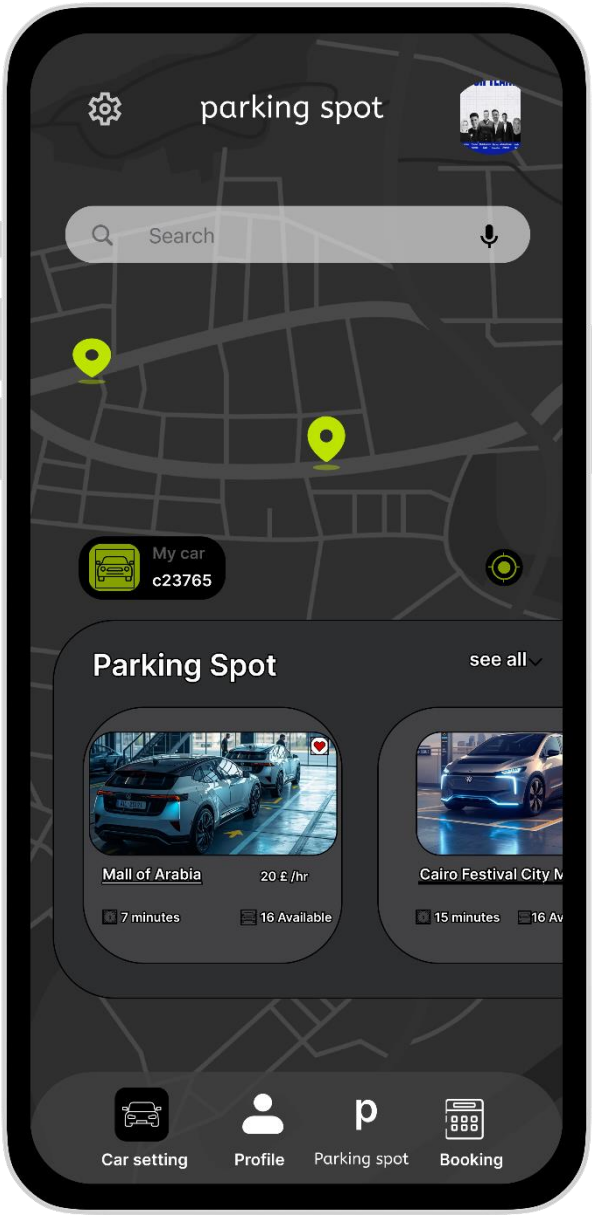
Verify Screen



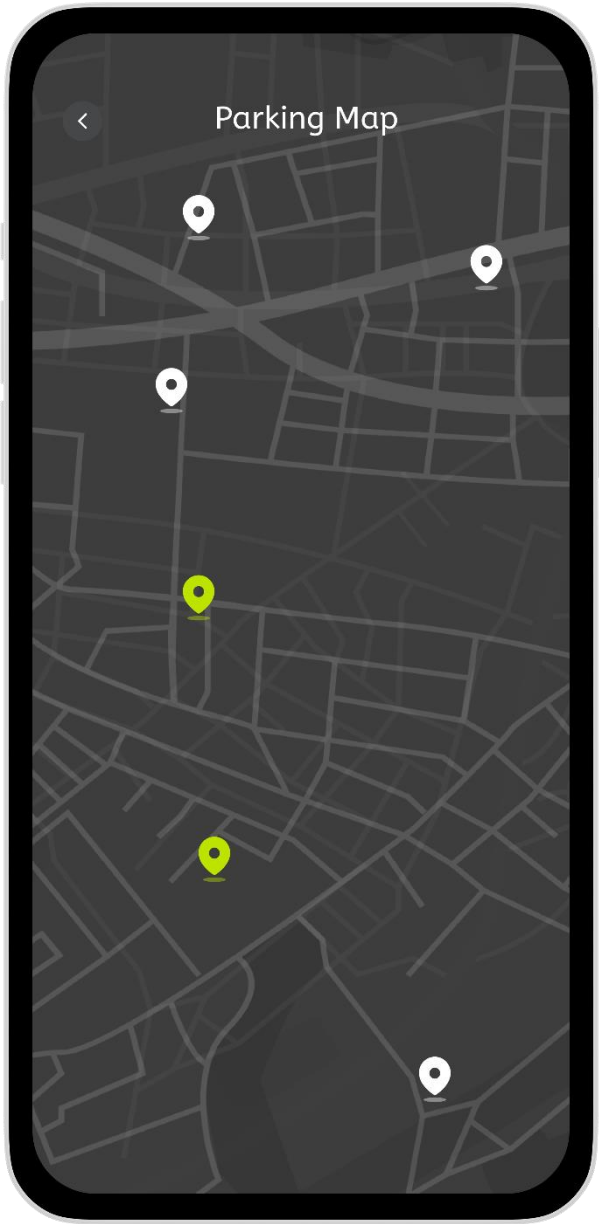


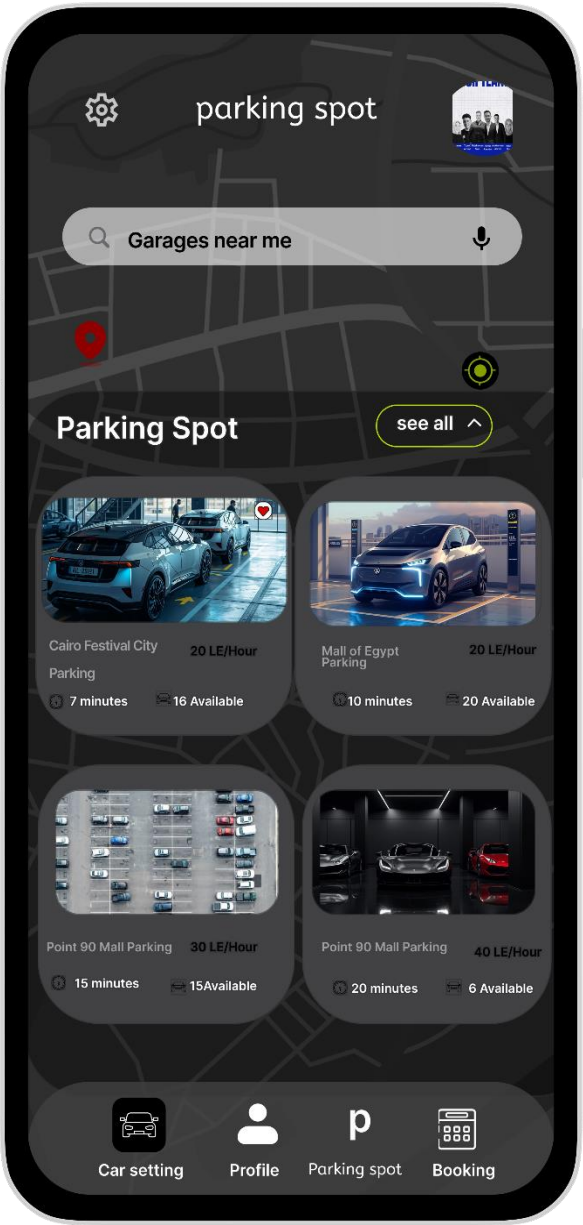
OTP Screen **OTP Enter Screen**



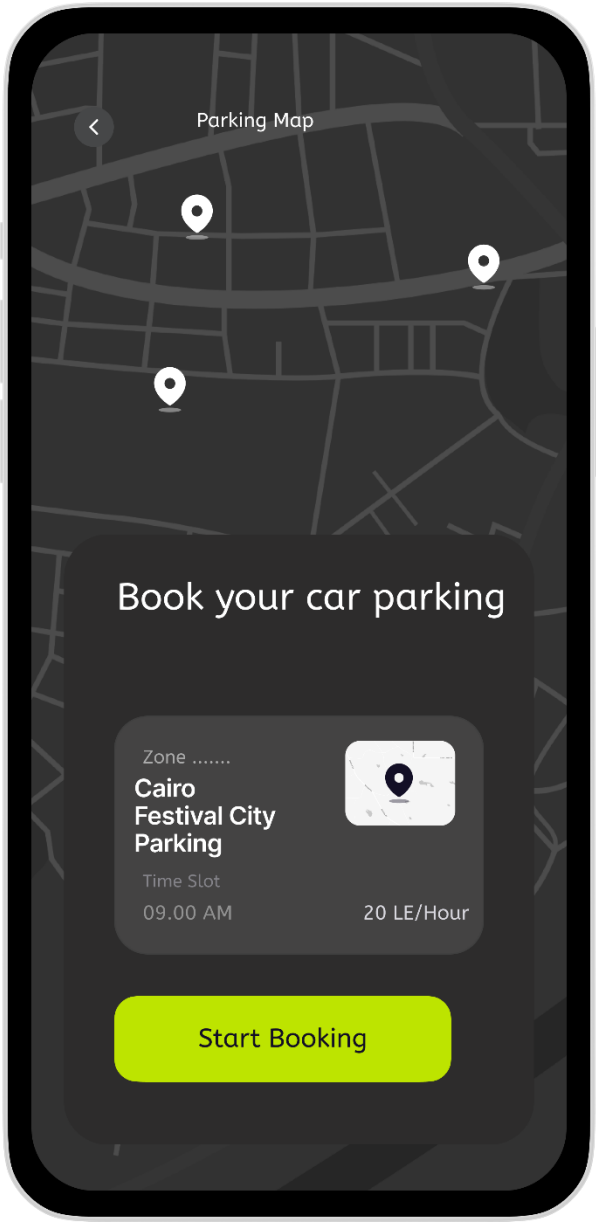


Verified OTP Screen **Home Screen**

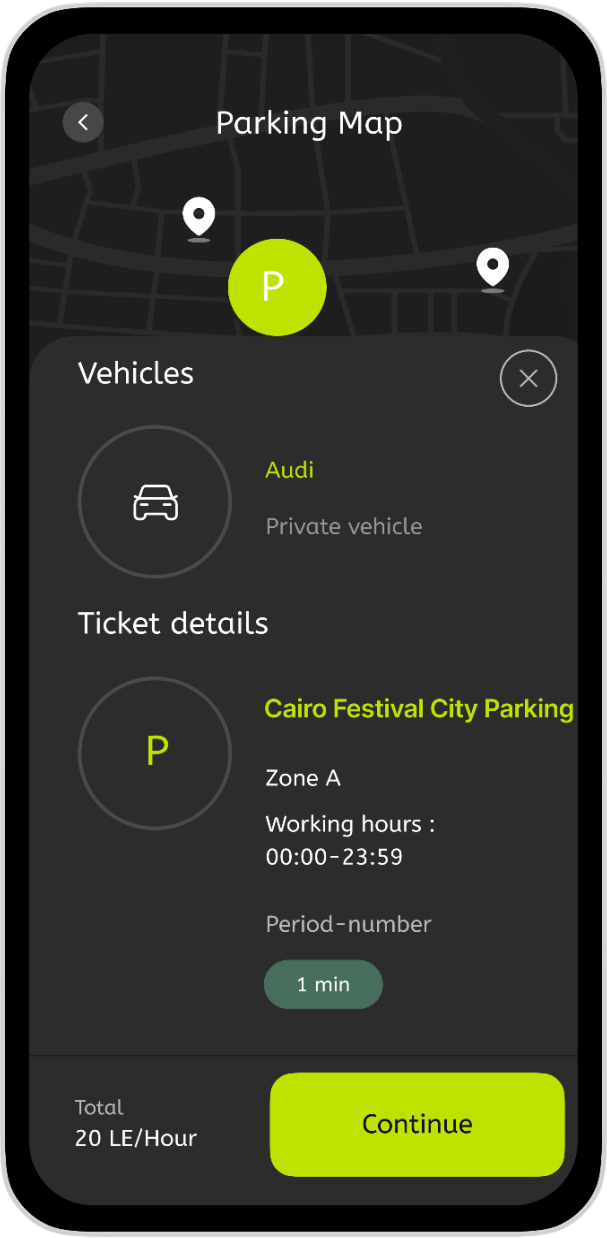




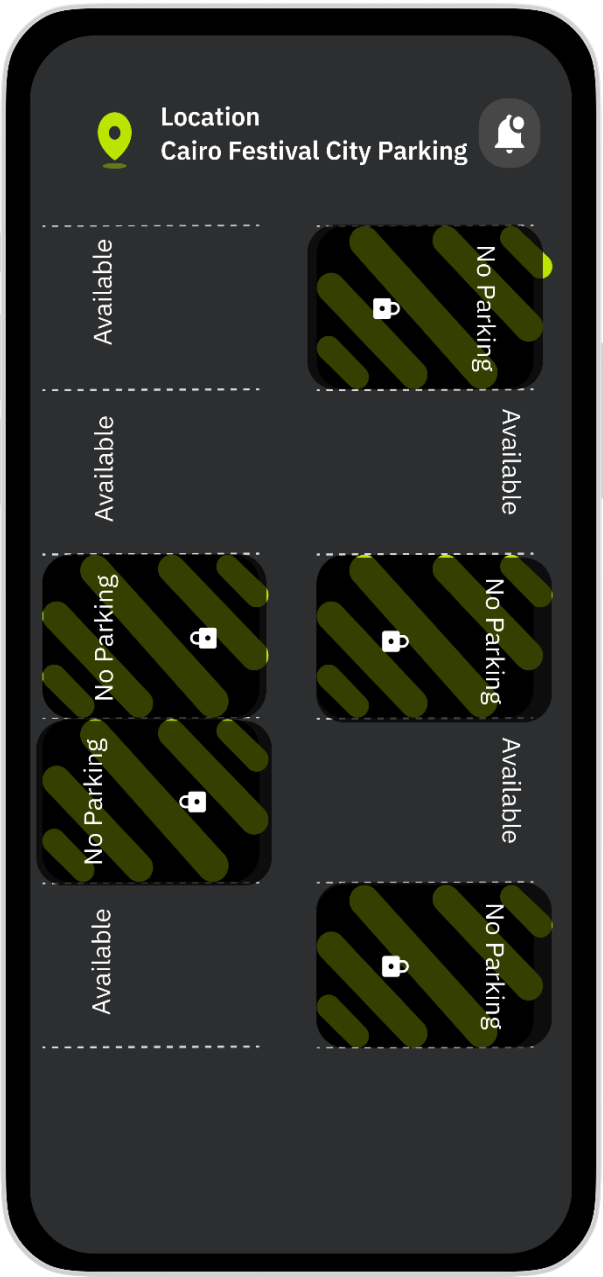
Map Screen **Parking Spots**



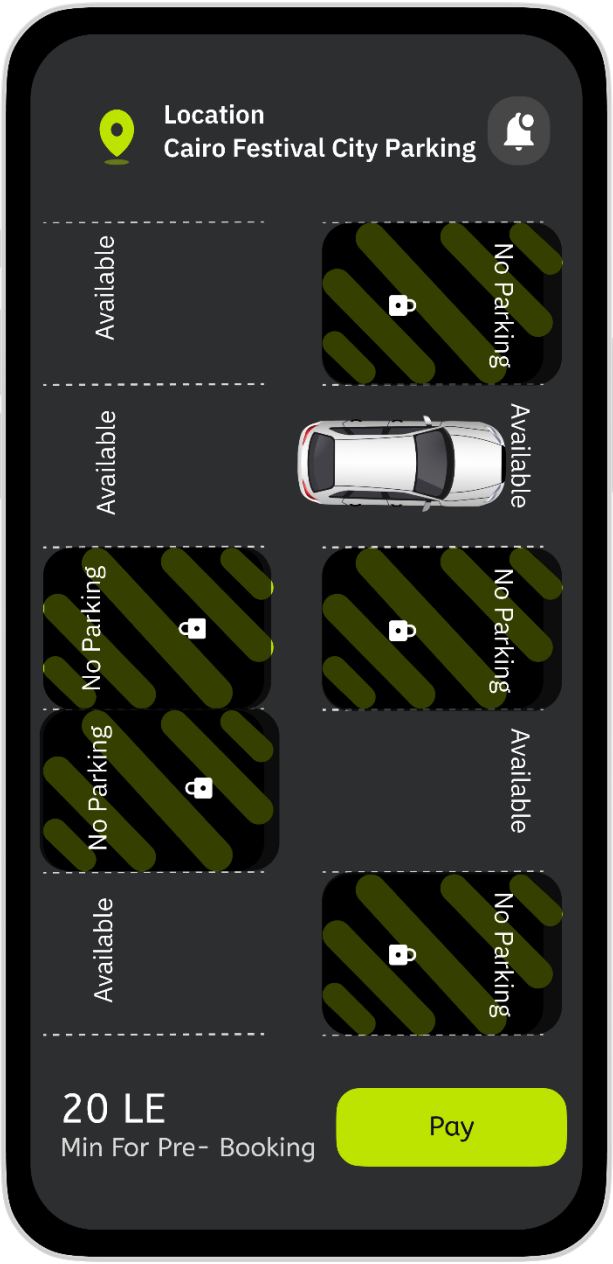
Booking Screen



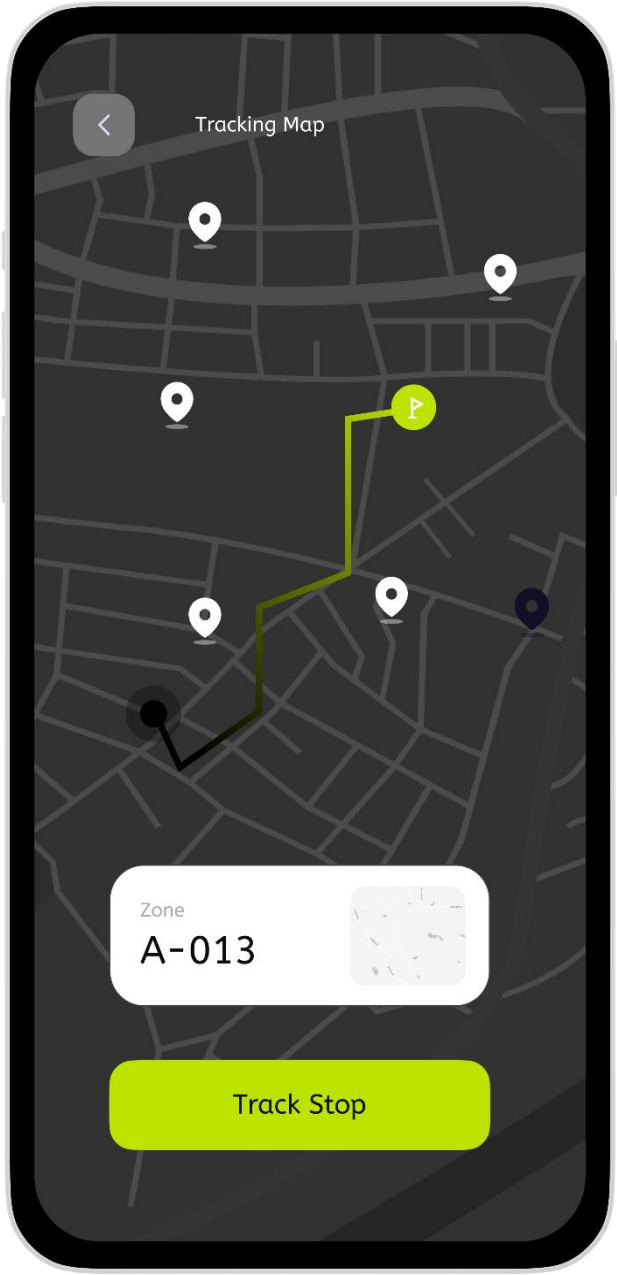
Continue Booking

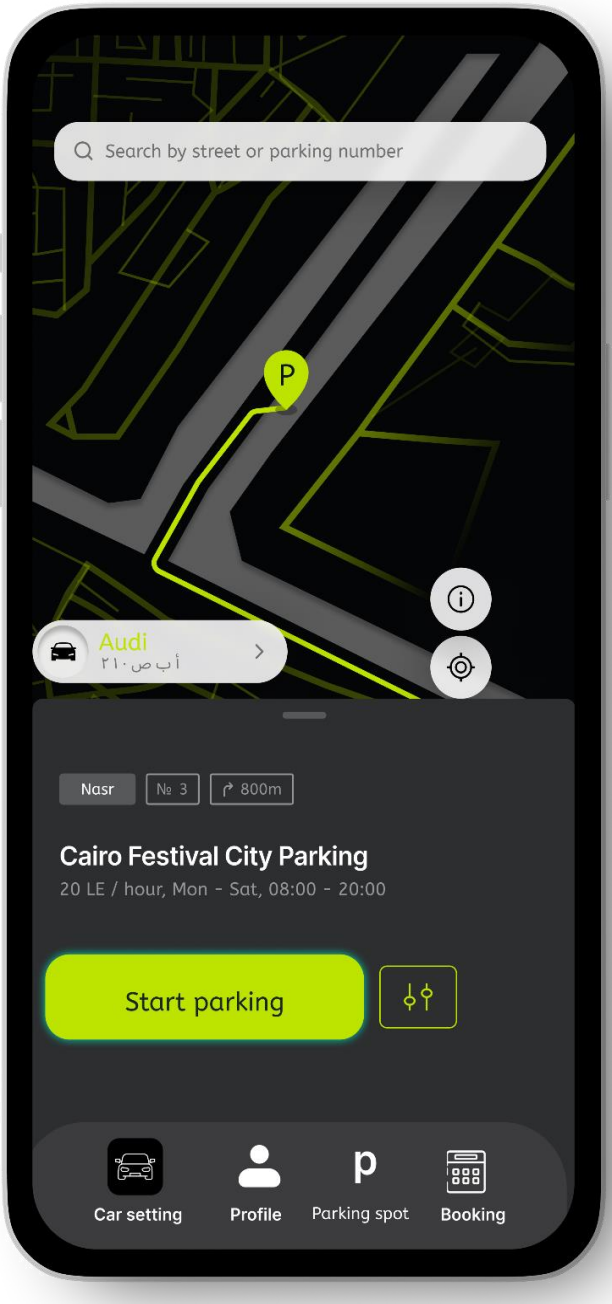


Choose Slot Screen

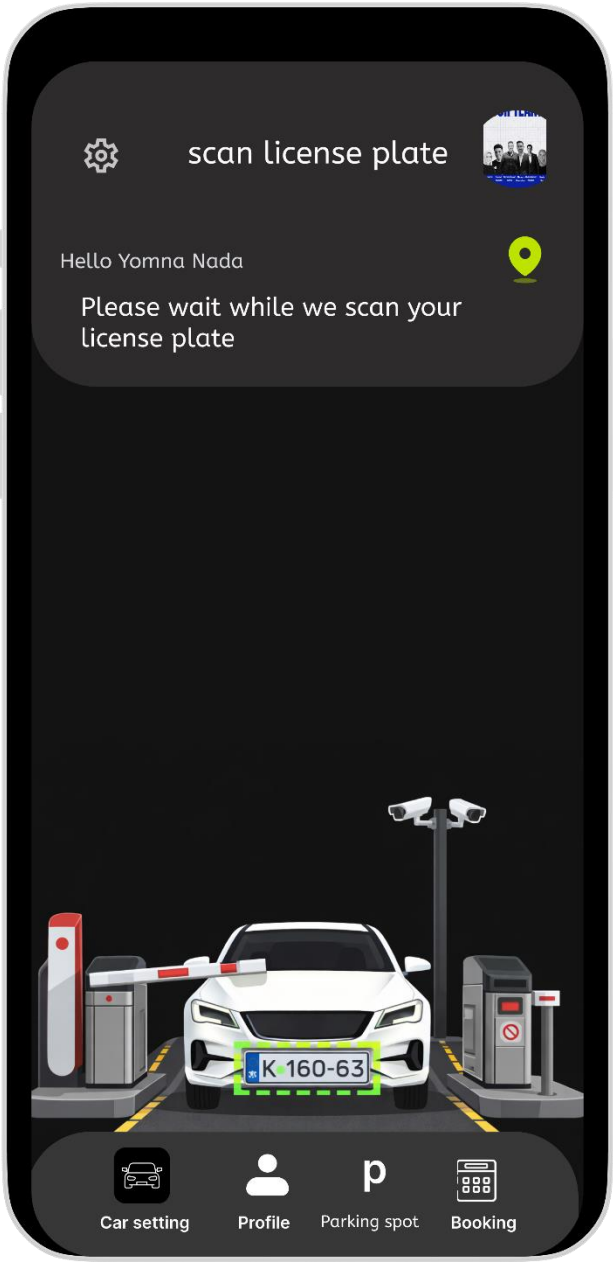


Pre-Pay Screen

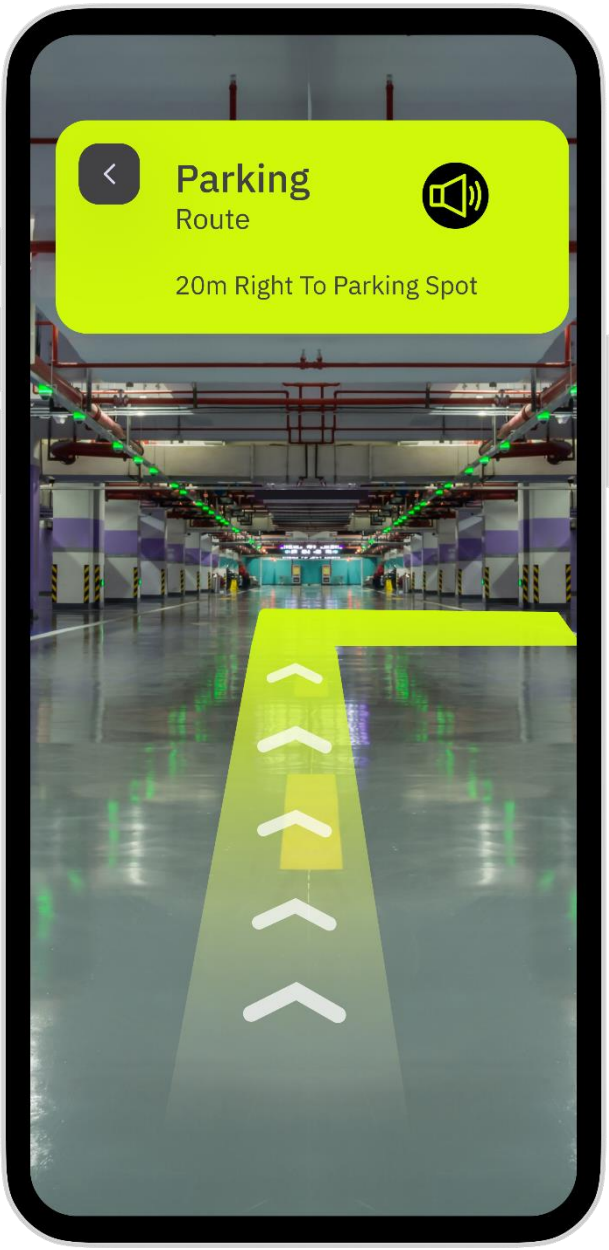




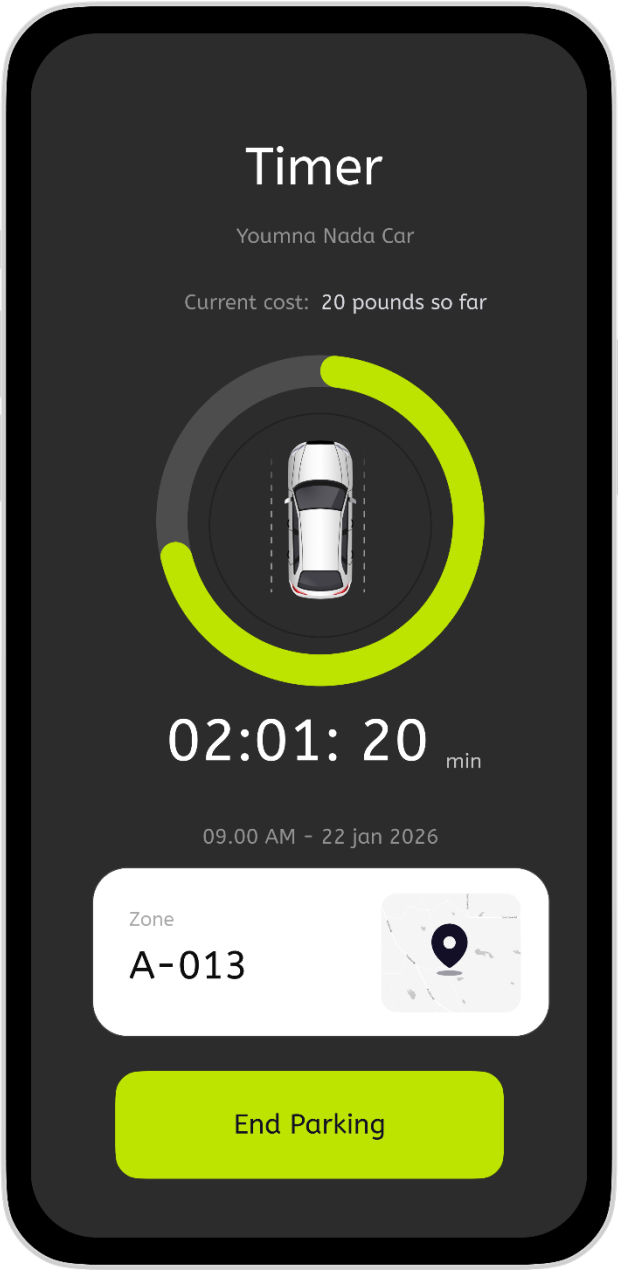
Track Slot Screen **Start Parking Screen**

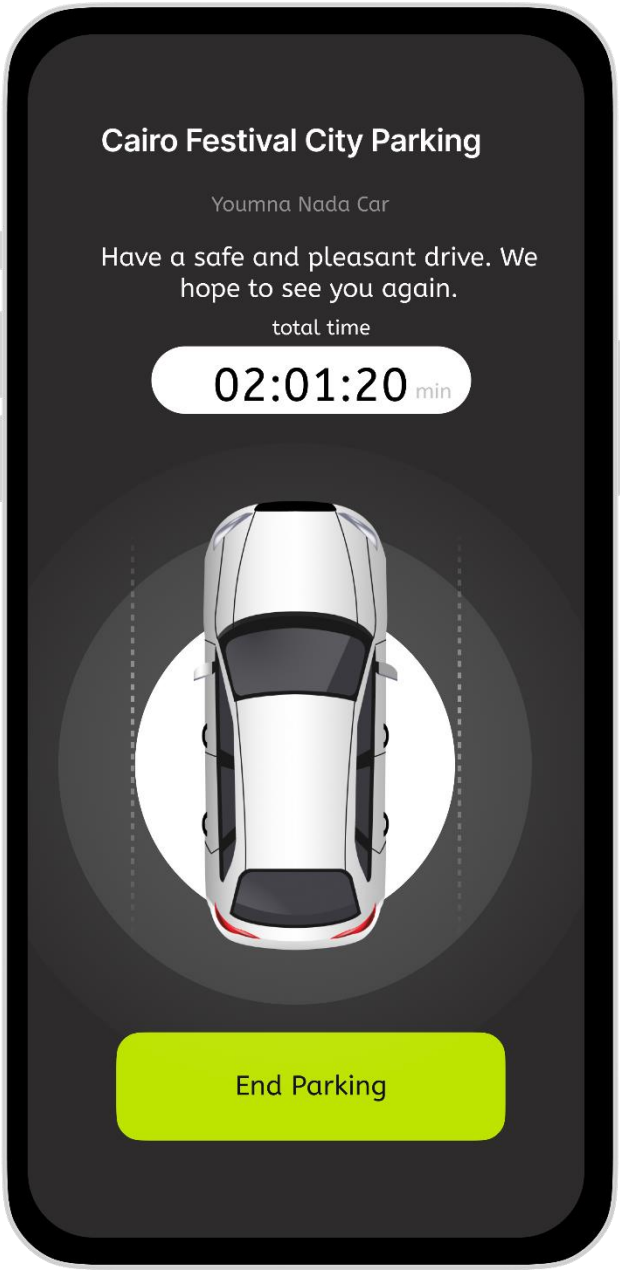


Scan License Screen



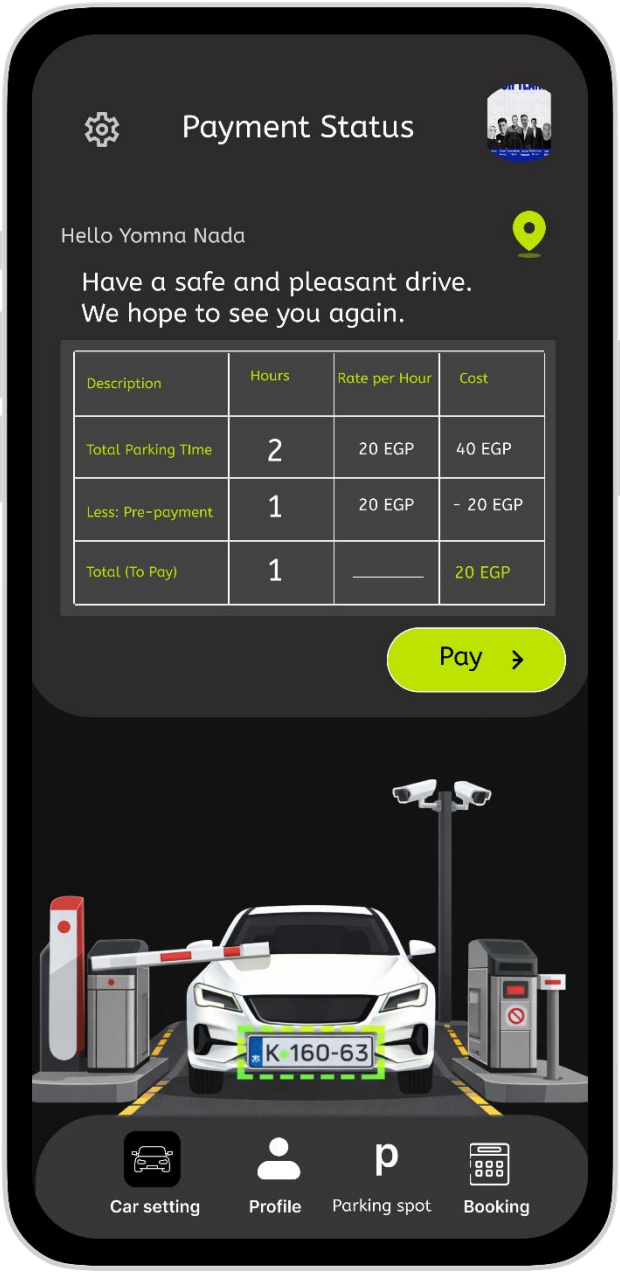
Navigate Slot

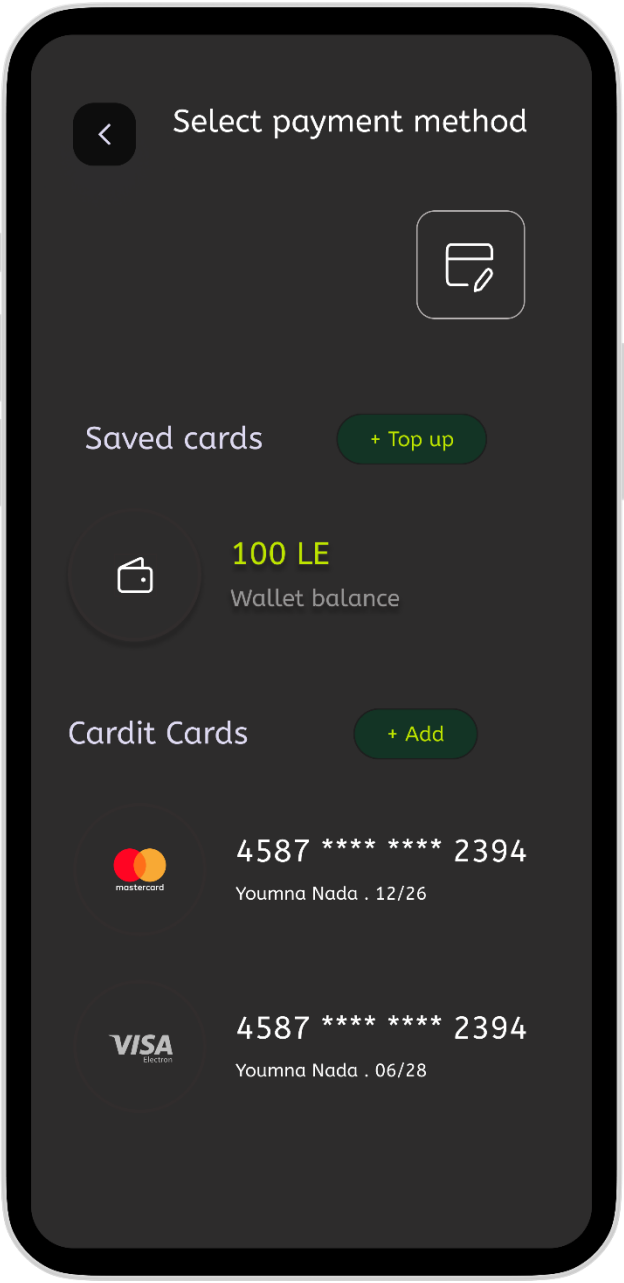




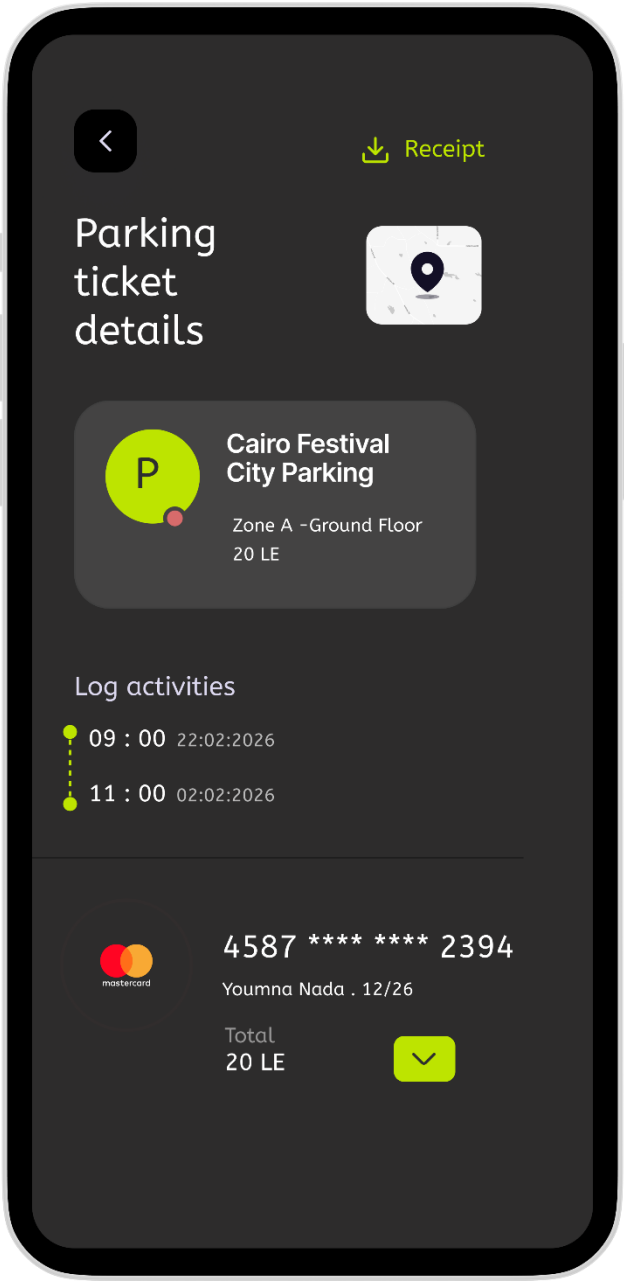
Time Screen

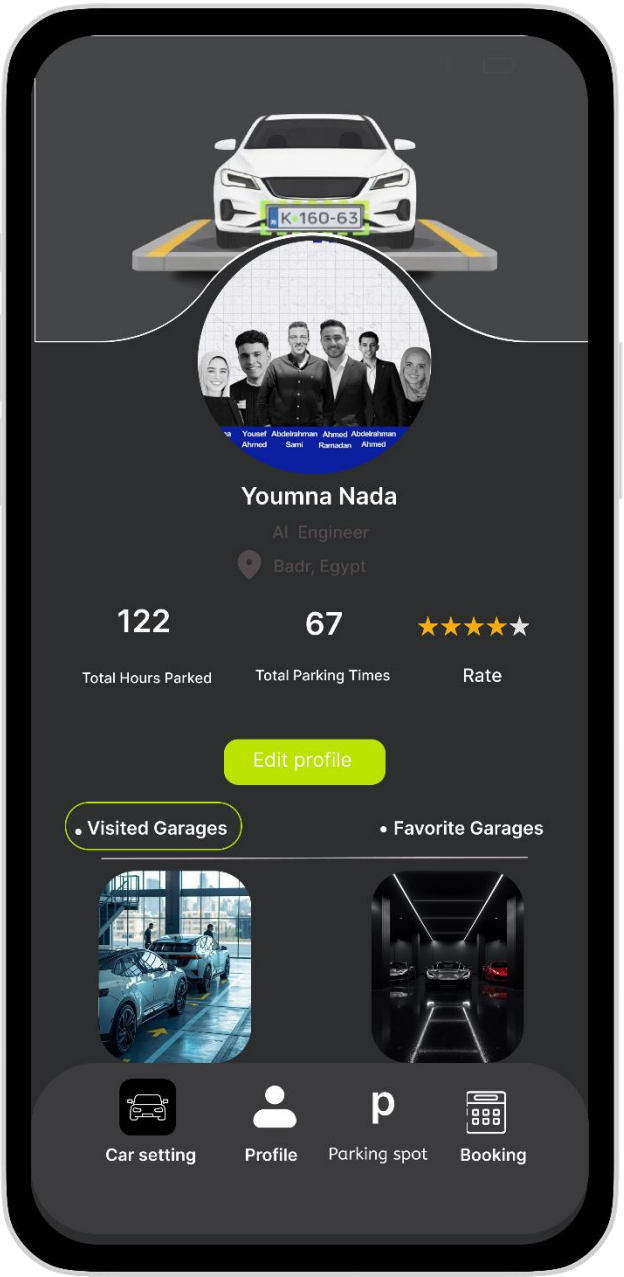
End Parking Screen





Cost Screen **Payment Screen**





Payment Screen **Profile Screen**

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